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Innovative Data Modeling Concepts for Big Data Analytics: Probabilistic Cardinality and Replicability Notations

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ABSTRACT

The evolving practice of big data analytics encompasses the aggregation of data from multiple sources, with the imperative of delivering metrics and reports that maintain a high standard of reliability and consistency. As stakeholders may interpretat the data and associated metrics differently throughout the process, this often have to make assumptions, which can lead to inconsistencies in metrics aggregation. Our work addresses the limitation of traditional data modeling methods, which often fail to capture the nuances of the relationships among various data sources. We propose two conceptual data modeling concepts: probabilistic cardinality and metric replicability along with definitions, notation and illustrative examples, as well as the general big data analytics framework that is used for discussing the role and implementation of the concepts. Application of proposed concepts is illustrated through two applied case studies highlighting variety of ways in which they reduce risk of inconsistent aggregation and reporting of metrics.

1. Introduction

With the growing availability of massive and diverse data sources, big data analytics has become an essential approach to turning information into knowledge in organizations. In contrast to traditional approaches, big data analytics leverages large volumes of heterogeneous and unstructured data [1], which can be used in isolation or combined with structured data originating from enterprise systems. The process of integrating the data from different information sources remains an open issue in big data analytics research, as the issues related to incorrect insights are often traced back to the way data is gathered, verified, stored and used [2]. The broad scope of issues in big data analytics has led researchers to examine the field across a variety of themes and methodologies. Some studies have focused on sector-specific applications such as health sector [3], the oil and gas industry [4], digital marketing [5], insurance industry [6], or transport and logistics [7]. Others have adopted a broader perspective, focusing on areas such as addressing research categories around big data analytics [1], developing a research framework for strategic business value creation [8], giving an overview and categorizing big data technologies/tools [9], classifying big data analytics literature [10], explaining mechanisms through which big data analytics leads to competitive performance gains [11], identifying key thematic areas of research on big data analytics in enterprises [12], etc.

Our work is motivated by the challenge of integrating the data from diverse information sources, whereby our proposed innovations are implemented during the conceptual data modeling stage of big data analytics. The semantic modeling additions we propose are designed to be applied during the data modeling stage of the big data analytics in organizations of any size or type, regardless of industry. Furthermore, conceptual level modeling is technologic-agnostic [13], meaning it does not depend on a specific implementation technology and can subsequently be translated into different logical and physical models aligned with an organization's specific data management platform and system capabilities. In addition, we recognize that the process of big data analytics varies significantly in practice and therefore we provide a general big data analytics framework that reflects the differences in the key steps of the process that can be taken. Additionally, we use this framework to discuss the proposed concepts in relation to the different approaches to the general steps of big data analytics outlined in the framework.

The main contributions of this study are as follows.

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- We develop a general big data analytics framework that integrates conceptual data modeling into the analytics process, capturing variations in real-world practice.
- We propose two new semantic data modeling concepts: replicability and probabilistic cardinality notation to address data integration issues arising from heterogeneous sources.
- We demonstrate the practical utility of these concepts through comparative case studies, showing how they improve metric consistency and model interpretability.
- We discuss the implications of these contributions for both data governance and future research in conceptual modeling for big data analytics.

2. Literature

2.1. Big data and database design

Organizations across nearly all industries today increasingly operate under the imperative of data-driven decision making, with decisions often relying on insights derived from big data. While there is no full consensus on the definition of big data, the most common set of concepts describing it is known under the abbreviation "3 Vs" which was first introduced in the 2000s by Doug Laney. The 3 Vs stand for volume, velocity and variety of the big data [14]. Generally, the term big data is primarily related to the idea of large volumes of data [15], but the 3 Vs definition was expanded beyond volume, variety and velocity by adding additional attributes such as veracity [16], value [17], variability [18], complexity [19], etc. Looking for a universal definition of big data is difficult, if not impossible and a suggested approach is to embrace all the alternative definitions instead, as each one of them focuses on a specific aspect of big data [20]. Beyond the volume of the big data itself, the big data revolution is characterized by technologies that capture time and location stamped data attributable to an event that used to have minimal digital trail such as a car trip, web searches and purchases, photos taken at social gatherings, postal tracking services and other activities [21].

When addressing the challenges and opportunities of big data and database design, it is important to consider current common practices. The foundation of all information systems consists of databases [22] and even though big data has disrupted the industry with innovative cloud solutions and information management systems, the most widely used database management system models are still relational databases [23]. Consequently, the most widespread data management technology are relational database management systems (RDBMS) [24], which provide the most understandable format for business application data while ensuring consistency properties required in business [25]. Data warehouses are central repositories for structured data integrated from diverse sources with architecture optimized for querying and analysis [26]. In addition to the various structured data sources that can be stored in data warehouses, organizations are managing ever-increasing amounts of non-relational and unstructured data coming from various data sources and/or recorded through various enterprise systems. Storage repositories that contain a mix of structured and unstructured data in their native format without pre-processing or modeling applied are called data lakes [26]. The growing demand to derive reliable insights by integrating structured and unstructured data poses a variety of managerial and technical challenges for organizations. In addition to standard data warehousing ETL (extract, transform, load) challenges, organizations are dealing with highly unstructured big data repositories and transformation of those repositories to structured data formats needs to be carried out in accordance with the analytics and metrics to be designed [27]. In simple terms, the process of getting data into the data warehouse transforms and optimizes the data for quick and efficient querying and analysis whereby designing and deploying the transformation process can be time-consuming and labor-intensive. On the other hand, the process of getting data into the data lake is quick, while the process of querying and analysis is more complex and requires a significant amount effort put into preprocessing and organizing the data before analysis.

In practice, the big data integration problems and the types of data involved in integration can vary considerably. One example of an organizational big data analytics process that combines structured, unstructured and semi-structured data sources in the healthcare industry context is an organization using electronic healthcare records as structured data source, logs of health monitoring devices as semi-structured data source, and clinical images as unstructured data source [28]. Another example comes from the streaming services industry, where structured data includes user profiles, ratings and subscription details, unstructured data consists of videos, subtitles and media content, and semi-structured data includes log files, smart TV data and social media interactions [29].

Finally, data integration is fundamental to big data infrastructure [30]. The distinguishing feature of organizational data that our work focuses on is enabled by big data capturing technologies. In simple terms, this feature arises from a phenomenon in which multiple instances of a particular event participant type or feature are attributable to a single event, whereas previously there was only one. This situation requires careful data integration, and this presents an opportunity for more refined insights based on a higher level of data granularity.

2.2. Organizational big data analytics

Big data, when considered in isolation, holds potential value for organizations [31]. This value can be realized through the process of Big Data Analytics (BDA), which transforms big data into meaningful and actionable insights. In order to successfully implement BDA process, the organization needs to have both adequate technical infrastructure and skilled human resources capable of designing, implementing and managing different parts of the process [8]. Furthermore, organizations need to have clearly defined business objectives as the BDA initiatives lacking clear business goals and strategies are likely to fail [8].

While there are different definitions and conceptualizations of BDA capabilities in the literature depending on research context, in simple terms, these capabilities refer to organizational ability to successfully implement and maintain the various aspects of the BDA process. Researchers have studied BDA capabilities in different domains with significant increase in number of contributions in the domains of general management, supply chain management, and health care [32]. In terms of organizational requirements for the BDA process, researchers have emphasized the importance of managerial expertise, data-centric business approach and organizational culture, and fostering organizational capabilities in addition to the technical and analytical expertise [33]. Regardless of the BDA capabilities research generally resulting in similar findings across different domains, the different antecedents highlight distinct resource priorities in different contexts [32].

The process of extracting knowledge from big data can be viewed as consisting of two main sub-processes: data management and analytics. Data management involves processes and supporting technologies to gather, store, prepare and retrieve data for analysis and analytics that involves various techniques used to analyze that data [34]. Regardless of the organizational context, there are general best practices that apply to the overall management of a big data environment, many of which are extensions from the DW/BI principles [35].

One general recommendation for organizations is to plan for disruptive changes on different fronts, such as emergence of new data types or programming approaches, and to anticipate the need to reprogram and rehost all big data applications within two years [35]. Therefore, in order to be ready for such disruptive changes, organizations should consider metadata-driven codeless development environment that can help insulate from technology changes [35].

The analytics process of extracting knowledge from big data can be classified according to the type of data analytics applied. The four main categories of data analytics are summarized in Table 1 below, along with

Table 1
Summary of data analytics types (adapted from Ref. [36,37]).

Data Analytics Type	Question Answered	Description
Descriptive	What happened ?	Simplest type of analytics used for summarizing and describing past data.
Diagnostic	Why did this happen ?	Identifies root patterns and trends observed in descriptive analytics by comparing coexisting trends and uncovering correlations.
Predictive	What might happen in the future ?	Used to make predictions that help with formulating strategies based on likely scenarios on future trends by analyzing historical data.
Prescriptive	What should we do next ?	Considers all possible factors and suggests takeaways for data-driven decisions. Goes beyond prediction to provide recommendations for optimizing future actions based on insights derived from previous.

general questions each addresses and a brief description of each.

Different organizations will select the type of data analytics that best fits their needs, and this choice will vary according to their strategic objectives, resources, and operational context. In most cases, the BDA process involves some form of data integration. The true value of this process lies in integrating and unifying diverse data sources, including existing data stored in legacy systems and other data sources that were collected at different time intervals, which provide varying levels of granularity and aggregation [8]. This additional data complements legacy system information by offering a more comprehensive understanding of the event such as retail transactions, patient visits, and manufacturing operations.

The issue of data integration in big data analytics has attracted interest across multiple research streams. For instance, research on big data analytics in the context of data warehousing focuses on transforming data stored in homogeneous sources into structured and interpretable formats [27]. Similarly, a major open issue in big data and predictive analytics research is the integration of data from multiple data sources to develop more accurate predictive models [2].

Moreover, organizations have recognized that seemingly tedious data problems are fundamentally business problems and cannot be resolved by IT teams alone. As a result, data governance has received growing attention from both practitioners and academics as a promising approach to addressing organizational data issues [38]. Big data governance is increasingly viewed as an ongoing process that must continuously adapt to meet the challenges of data-driven organizations [39]. One such challenge that big data introduces is the need to enforce governance principles even when stakeholders lack clear expectations of the data content, with a key recommendation being to dimensionalize the data as early as possible since the dimensions provide the “who, what, where, when, why and how” context surrounding a business process event [35]. In other words, all related data representing possible classifiers from different sources should be organized in a way that produces groupings of metrics in a consistent manner.

Following that recommendation, our work focuses on those specific aspects of big data governance through a more comprehensive approach for modeling many-to-many relationships. The objective of adding new, relevant semantic components to existing modeling techniques used in practice is to support strategy and data governance while reducing risks associated with inaccurate reporting caused by traditional modeling approaches that often fail to capture the nuances of relationships illuminated by big data. While general challenges posed by big data are frequently addressed from a technical perspective, it is equally important to address them from a management perspective. For this reason, our proposed methods are presented through the discussion of new semantic concepts, illustrated by examples and discussed in context of the overall process. Our approach is deliberately high-level and does not require novel expertise, as we employ traditional data modeling techniques to explain and discuss the proposed concepts.

2.3. Conceptual data modeling

Data models differ in complexity and level of detail and are typically categorized as conceptual, logical, or physical data models. While the focus of this work relates to the conceptual data modeling, we briefly provide descriptions of all three as per [15]. Conceptual data models represent high-level, platform-independent views of the data problem domain through entities and relationships. Logical data models refine conceptual models while remaining platform-independent, including details such as unique identifiers, primary keys, foreign keys, and additional relationships. Physical data models are platform-specific and used during implementation, where previously defined entities and attributes are represented as tables and columns of specific data types that reflect platform-specific aspects.

Conceptual data models are often used to present structured business view of the data that supports business process, records business event, or track performance measures [40]. Conceptual data models can be developed at varying levels of abstraction, ranging from high-level overviews to detailed, granular representations of data generation and consumption processes. In general, a conceptual model can be described as an abstract and simplified view of the world that represents a specific area of interest, with its most important quality characteristic being ease of modification and adaptability [41]. The simplicity of conceptual models makes them easy to understand by both technical and non-technical users [42]. Moreover, the process of conceptual data modeling is no longer limited to technical stakeholders. Although conceptual models have traditionally been used by IT experts for the purpose of designing databases, engaging non-technical stakeholders to develop and use conceptual models can be beneficial [43,44].

In the field of database design, the Entity-Relationship (ER) model was originally developed in 1976 by Peter Pin-Shan Chen [45] and remains to be one of the most commonly used tools in the conceptual data modeling process. Its continued prevalence in database design can be attributed to its simplicity, intuitive appeal, and ability to capture useful semantics across different domains [46]. Grounded in set and relation theories, the ER model adopts the natural view of the real world consisting of entities and relationships, providing a logical view of the data [45]. ER modeling effectively describes even a less structured domain in an understandable manner, is widely taught in academia, and easy to use [22]. Using ER modeling to define data structure is more attractive to users and requires less effort than learning a new conceptual model [22]. A systematic literature review on design methods for the new database era conducted by Roy-Hubara and Sturm [22] found that the ER modeling is still the most commonly used conceptual modeling approach among researchers, regardless of the design method implementation of choice [47–54].

As the modern applications require capturing and enforcing more domain semantics than traditional applications, Badie [46] proposed for these semantics to be captured through conceptual modeling extensions. The recommendation was that such extensions should be well-motivated, minimal, conceptual and powerful. Our work follows this recommendation, as our data modeling extensions fulfill those requirements. We present our concepts as extensions of the Entity-Relationship (ER) modeling notation as it is the most commonly used modeling language for the conceptual level data modeling [55]. Furthermore, in big data modeling research, ER modeling is considered a prevailing trend for data modeling at the conceptual level [13], and our approach aligns with current research directions. While our proposed notation is compatible with the relational implementation of ER modeling, it is not limited to it and can be applied using other data modeling approaches. To emphasize its broader applicability, we introduce a general framework for big data analytics in the following section, which enables the notation to be interpreted and implemented across various modeling methodologies.

Finally, the process of conceptual modeling is a demanding task that aims to represent aspects of the reality in an objective, understandable, shareable and reusable form, thereby providing a shared foundation for organizational stakeholders to support software system deployment and development or to formalize the domain understanding across different processes or tasks [56]. For the conceptual model to remain accurate, it needs to be continuously updated to reflect business needs. In other words, the conceptual data modeling process includes both the initial creation of the model and its ongoing maintenance. This process facilitates communication and understanding among stakeholders, promotes shared vision of complex data, and ensures data consistency across systems. Conceptual data modeling promotes a shared vision and a common understanding of the domain information, facilitating interoperability with other systems in unforeseen ways at development time [41].

2.4. Related work

In this section we provide a summary of recent related research publications along with a short description of their work and the abstraction level of the data models in their work (Table 2). We highlight interplay of conceptual, logical and physical models for each of the related works to contrast it with our approach.

We intentionally selected a diverse range of published work to illustrate the variety and scope of contributions in research areas related to our study. The related work summarized in Table 1 above presents a wide range of approaches and methods that applied to different problem contexts that fall under the broad category of data integration issues, whether these challenges arise from operational or analytical needs of organizations. All contributions can be described in terms of their level of data modeling abstraction, as they address distinct stages of the data modeling process. For instance Ref. [58], emphasize the physical level [60,61,63], concentrate on the logical level, and [62] address the conceptual level. Several studies propose methods or models designed to support transitions between different levels of abstraction. For example [59], address the transition from physical to logical (reverse engineering, in this case justified) [57], from physical to conceptual, and [53] from conceptual to logical to physical levels. Our proposed concepts are primarily positioned as contributions to conceptual modeling. However, we extend these concepts to the logical and physical levels through illustrative examples, thereby demonstrating their applicability across all three levels of abstraction.

Most of the related work focuses on particular type of system, with proposed solutions tailored to the context of that system. For example [53,57,60,61], present contributions related to NoSQL systems. [58], in turn, propose a model that concentrates on heterogeneous data sources, although their approach is not general, but specific to healthcare domain. In contrast [62], introduce ER+, an extension to EER model that is independent of the context, technology or domain. The ER + aims to provide developers with a unified modeling framework capable of representing distributed data architecture, data transport and data transformation. Our approach is intentionally designed to be independent of specific contexts, technologies, and professional roles. To simplify the complex process in which our concepts are applied, we introduce a big data analytics framework. This framework allows the BDA process to be understood in a generalized manner rather than being constrained to a single system or approach.

Finally, our concepts are positioned for application within the broader context of big data analytics. Although we do not present highly complex end-to-end implementations, our approach demonstrates processes at a detailed level, highlighting multiple alternative workflows that reflect real-world analytics practices, in which no single implementation method is universally applicable. The rationale for our approach lies in deficiencies frequently observed in day-to-day practice, which have emerged with the advent of big data technologies and the resulting qualitative and quantitative shifts in data. These shifts have altered data representation, evolving from aggregated and simplified forms to highly granular records, thereby creating new challenges for modeling, design, and deployment.

Table 2
Big data modeling: related work and contributions.

Author	Contribution/Focus	Abstraction Level
[57]	“ToConceptualModel”: proposed automatic process model for transforming physical model into conceptual to make it easier for developers and decision makers to understand database and write queries in NoSQL databases.	Physical → Conceptual
[58]	Proposed a model focusing on ontology-based data integration method for data types: excel, SQL and MongoDB(NoSQL). Used XML-based ontology creation and mapping (XML global schema).	Physical
[59]	Introduced a metadata model to support semantic annotation and profiling of multidimensional data. Graph representation for metadata at source and attribute levels. For SQL, JSON/XML files unified by URIs (uniform resource identifiers).	Physical → Logical
[60]	“SRank”: proposes a new approach to analyze and evaluate different schema alternatives and suggest the best schema out of various NoSQL schema alternatives. SRank helps to assess how well a schema and query work together.	Logical
[53]	Propose a methodology for obtaining a detailed design from a conceptual model. Phases of proposed NoSQL document database develop consists of high or conceptual level (conceptual model and access patterns), logical level (types of documents, interrelations and specifications), and physical design in steps like phases of traditional relational database. Query patterns and ERD for conceptual, Document integration diagram (DID) for logical: classes or types of documents, their structure and interrelation. JSONSchema describes structure of another document.	Conceptual → Logical → Physical
[61]	Propose NoAM (NoSQL Abstract Model) for NoSQL databases which exploits commonalities of various NoSQL systems. Propose a database design methodology for NoSQL systems based on NoAM, with initial activities that are independent of the specific target system. NoAM is used to specify a system-independent representation of the application data which can be implemented in target NoSQL databases, taking into account their specific features.	Logical
[62]	ER + notation for depiction data transformation and data transport between different locations of a system. They describe proposed symbols and demonstrate their use through examples. Proposed concepts are used for location definition, entity notation add-on, and functional relationships.	Conceptual
[63]	Design and implementation of “EverAnalyzer”, a self-adjustable Big Data management platform that selects optimal Hadoop ecosystems frameworks for different processing and analysis scenarios.	Logical

3. Big data analytics framework

An organization’s capacity to capture, integrate and analyze big data, and to utilize the resulting insights, constitutes a core dimension of big data analytics capability [64]. A central aspect of this capability is the ability to manage heterogeneous data sources throughout the analytics process, a challenge that underscores the importance of robust data modeling practices. Therefore, the management of different types of data that originate from various sources is essential for advancing big data analytics and enhancing its application [65].

Our proposed innovative data modeling methods and concepts are articulated through conceptual models, designed to be incorporated into the data modeling process and accessible to all stakeholders through the analytics lifecycle. Conceptual models provide a common language for stakeholders with different backgrounds such as domain experts, business professionals, data scientists, and engineers, and they support proper algorithm use by grounding data-driven decision-making in a clear conceptual perspective [66].

To contextualize these contributions, we introduce a framework that structures the deployment and utilization of big data analytics in organizations as shown in Fig. 1. The framework begins with the **definition of business case and requirements** where domain experts establish objectives, domain-specific activities, and stakeholder responsibilities, while also considering constraints such as budget, internal capabilities, and time.

The next component, **data sources**, encompasses three broad categories: unstructured big data, relational databases, and NoSQL systems. Big data typically refers to the high-volume, high-velocity, and high-variety unstructured data originating from various sources. Relational databases, by contrast, capture data in structured formats with predefined schemas, while NoSQL is widely used as an umbrella term for non-relational databases [67], which include column-oriented, document-oriented, graph-based or key-value pair databases [68].

The **data modeling** stage can involve a variety of data modeling techniques, though in some cases no explicit modeling is deployed at this stage. If the modeling is conducted in this stage of the framework, it is labeled as the design phase of the "schema-on-write" approach. Relational databases are grounded in the "schema-on-write" approach, while NoSQL databases generally rely on "schema-on-read" where schema definitions are implicit or determined at the implementation stage [69]. Depending on the chosen approach and the organizational data sources, organizations may adopt extraction, transformation, and loading (ETL) into a data warehouse, or extraction, loading, and transformation (ELT) into a data lake. Hybrid strategies are also common, reflecting the specific characteristics of the data and analytical requirements. In the case of a data warehouse, schema-on-write designs are implemented as structured databases into which transformed source data is loaded.

The **analytics** stage varies in complexity, ranging from statistical analysis performed on a data warehouse to advanced methods such as machine learning and data mining. Under "schema-on-read", schema design and implementation occur at this stage in and ad hoc manner.

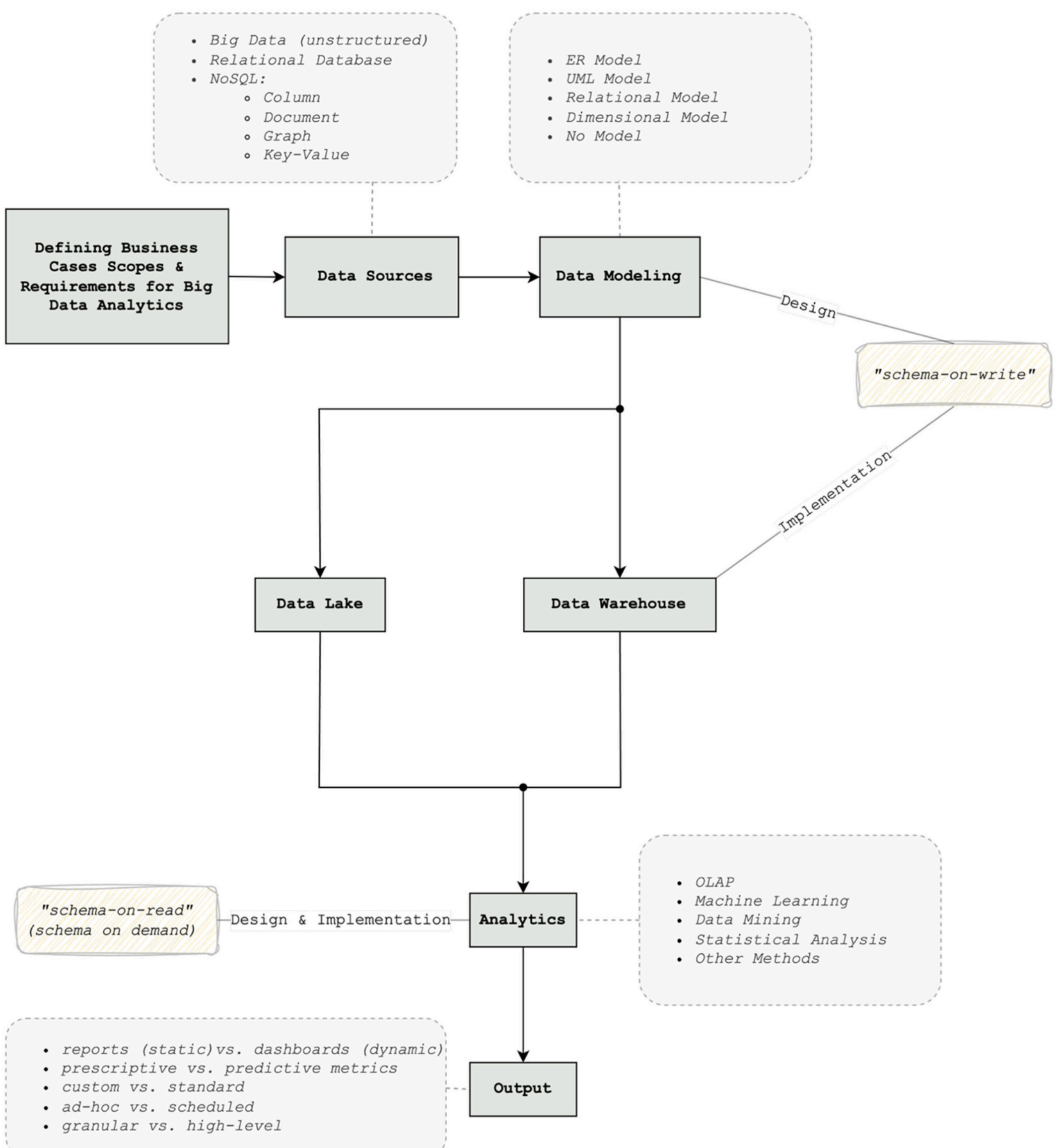


Fig. 1. Big data analytics framework.

Finally, the **output** stage generates results aligned with organizational needs. Outputs may include scheduled static reports or interactive dashboards, with granularity ranging from detailed metrics for individual stakeholders to high-level comparative overviews for executives.

As summarized in Fig. 1 above, the framework illustrates the interdependence among data sources, modeling choices, and analytical process. In big data research, innovative algorithms cannot compensate for the lack of sound modeling practices [70]. Therefore, our focus is on enhancing modeling practices by introducing semantic data modeling extensions for big data analytics in the following section.

4. Semantic data modeling additions for big data analytics

In this section we introduce and discuss our contribution of two novel semantic concepts, followed by illustrative examples. Both of these concepts address the need for more accurate and efficient metric aggregation and reporting as big data sources are increasingly being used in combination with traditional transactional data.

4.1. Probabilistic cardinality notation

We introduce our contribution through the lens of conceptual modeling and the ER diagram. As a starting point, we outline relevant background concepts and present simple examples that illustrate connections between multiple data sources at the instance level. These examples are deliberately simplified, yet the same principles apply to big data environments where many such instances coexist. To highlight the issue more precisely, we examine it at a finer level of detail.

Entities in an ER diagram can typically be divided into two types: transaction entities and component entities. Transaction entities represent events such as payments, transactions or reservations, while component entities are related to the transaction entity through a one-to-many relationship and they answer questions such as who, what, or where about that event [71]. For one-to-many relationships where the entity on the "one" side is a component entity and the entity on the "many" side is a transaction entity, the standard mapping of ER models into the corresponding dimensional model, and consequent apportioning of business analytics metrics, must be handled in a different manner than when such relationship is one-to-one.

A star schema, as a product of dimensional modeling, consists of a fact table that typically stores data mapped from transaction entities and serves as the central table surrounded by dimension tables containing data mapped from component entities [71]. The dimensions are often highly denormalized tables that include one or more embedded hierarchies providing the basis for data analysis [71]. Standard mapping between a dimension instance and a fact instance without apportioning metrics of interest is applicable if such relationship is one-to-one, whereas a method for metric apportioning must be applied when multiple transaction instances relate to a single component entity instance.

As the volume and granularity of big data increase, data relationships that were traditionally one-to-one or one-to-many are now more frequently observed as many-to-many. To address this shift, we propose a probabilistic cardinality notation that captures these relationship dynamics. We demonstrate how this notation can be integrated into the big data analytics workflow and discuss its practical implications.

In simple terms, the probabilistic cardinality notation addresses the varying probability that an instance of an entity on the one side of the one-to-many relationship will be associated with more than one instance of the entity on the many side. We provide a conceptual definition of the proposed probabilistic notation and describe it through the ER modeling approach to clarify the mapping process and its implications for ETL procedures. We contrast the proposed probabilistic cardinality notation with standard ER notation in Table 3. While this comparison uses one-to-many (1:M) relationship, the proposed notation can also be applied to many-to-many (M:M) relationships, which we further demonstrate through an illustrative example in Section 5.

Table 3
Probabilistic cardinality notation summary.

Notation	Cardinality	Description
Probabilistic Cardinality Notation (1:m)	(1, m)	This notation indicates that the probability that the entity on the "1" side of the 1:M relationship is associated with more than one instance of the entity on the "M" side must be evaluated. Traditional "M" side in this case is denoted as "m" and depending on the evaluation of the probability, "m" can further be treated as "1", as "M", or as two separate relationships where: primary treated as "1" and secondary treated as "M".
Standard Notation (1: M)	(1, M)	The standard one-to-many relationship in ER modeling mandates that even if the minimal likelihood of one entity being related to more than one instance of another entity exists, the relationship must be described as one-to-many.

In probabilistic cardinality notation, m indicates that the relationship should not automatically be treated as a standard 1:M. At the conceptual modeling stage, the relationship may be expressed generically as $1:m$. During implementation, however, the abstract placeholder m is replaced by an actual cardinality value of either $1:m(1)$, $1:m(M)$, or $1:m(1,M)$. The approach to choosing the actual cardinality value of m can be process-driven and/or data-driven.

To formalize the notation, we define R as a relationship type between entity sets E_1 and E_2. We define the cardinality constraint as

$$\mathrm{card}(E_1, E_2) = (1, m),$$

where

$$m \in \{\ m(1), m(M), m(1, M)\ \}.$$

The possible cardinality values on the m side of the relationship are:

- (1): exactly one,
- (M): many,
- $(1, M)$: one and many, represented as two distinct relationships.

In the case of cardinality value corresponding to (1,M), the relationship is decomposed as follows:

$$m = m(1, M) \Rightarrow R = R_{(1:1)} \cup R_{(1:M)},$$

with

$$R_{(1:1)} \subseteq E_1 \times E_2, R_{(1:M)} \subseteq E_1 \times E_2,$$

such that each $e_1 \in E_1$ participates in a primary $1:1$ association via $R_{(1:1)}$ and a secondary $1:M$ association via $R_{(1:M)}$.

The process-driven approach of choosing cardinality value for m focuses on the organizational process that caused certain instances (e_1) of E_1 being associated with more than one instance (e_2) of E_2 and the consequent frequency of such outcome. For example, certain instance (e_1) of E_1 being associated with more than one instance (e_2) of E_2 can be allowable outcome of a standard organizational process. Even if the probability of such outcome is low, this relationship is treated as a standard 1:M relationship. On the other hand, if observation of certain instances (e_1) of E_1 being associated with more than one instance (e_2) of E_2 is a result of a non-standard practice that occurs rarely but is permissible, then this relationship could be broken into two separate relationships. Finally, if an occurrence of such outcome is a result of rare exception to the standard business practice, the relationship may be treated as 1:1.

A data-driven approach may also be employed, in which the domain-specific threshold τ is derived from historical observations estimating how often an instance is associated with more than one counterpart. In order to express the likelihood of multiple participation in the relationship, we define $N(e_1)$ as the number of instances in E_2 associated with a given $e_1 \in E_1$. The event that e_1 exhibits many-side behavior is characterized by the probability

$$p(e_1) = \Pr\ (N(e_1) > 1),$$

which informs the interpretation of the cardinality term m within the probabilistic cardinality notation. Once the threshold is specified, the cardinality value of m is determined by comparing $p(e_1)$ to τ. To incorporate organizational or domain-specific exceptions, let the $E_{\text{exc}} \subseteq E_1$ denote the set of instances requiring special treatment. The resulting threshold rule is expressed as

$$m = \begin{cases} m(1) & \text{if } p(e_1) < \tau, \\ m(M) & \text{if } p(e_1) > \tau, \\ m(1, M) & \text{if } p(e_1) = \tau \text{ or } e_1 \in E_{\text{exc}}. \end{cases}$$

In this rule, the condition $p(e_1) = \tau$ maps to the mixed cardinality $m(1, M)$; however, this choice is not mandatory. Depending on the business context, the $p(e_1) = \tau$ case may instead be grouped with either $m(1)$ or $m(M)$. Conversely, if the probability of multiple associations is *too high* relative to what is acceptable for a 1:m(1) interpretation, the relationship is treated as 1:m(M); if it is sufficiently low, it remains 1:m(1). Certain relationships between entities nonetheless involve complexities not reflected in this probability comparison and must be handled as exceptions.

As the process of determining actual cardinality of m is repeated over time, the threshold may fluctuate to reflect changes in data and the business practices that generate them. The choice of a particular threshold value should therefore be tailored to the domain context and organizational policy. A summary of the approaches for selecting the actual cardinality value of m and their operationalization is provided in Table 4.

These two approaches are mutually supportive and complementary in practical implementation. In some cases, organization possesses a detailed understanding of their business rules and processes and therefore applies process-driven approach from the outset, with observed data patterns simply confirming threshold values that are already implicit in these practices. In other cases, organizations may rely on data-driven threshold calibration, where sustained use over longer periods uncovers previously unrecognized and unspecified organizational rules that can subsequently be formalized.

For the crow's foot notation in ER diagrams, we introduce a dashed-line symbol to represent probabilistic cardinality m, while retaining the solid line used in standard notation to represent standard cardinality M. This extension enables the explicit representation of uncertainty in relationship cardinalities. Fig. 2 illustrates the notation through a teacher–class example, where a relationship that would traditionally appear as M:M is now more precisely expressed as m:M, indicating probabilistic rather than guaranteed multiplicity on one side.

Table 4
Approaches to selecting actual cardinality value of ***m***

Approach Type	Description of selection of actual cardinality value of ***m***
Process-driven	• Determine organizational policy or domain rules that define acceptable multiplicity between entity instances. • Identify substitute or edge-case relationships where multiple links are permissible and set a policy-based rule accordingly. This approach formalizes policy-driven distinctions, such as treating rare but sanctioned 1:m outcomes as m(M), decomposing non-standard ones m(1,M), and classifying insignificant exceptions as m (1).
Data-driven	• Estimate $p(e_1)$, the probability that an instance $e_1 \in E_1$ is linked to more than one instance $e_2 \in E_2$, from historical data or observed cardinality ratios. • Tune τ periodically to account for data changes that are driven by process changes or domain drift. This approach represents calibration driven by objective data-driven domain-specific threshold values.

We illustrate our proposed notation in a scenario presented in Fig. 3. Here, the relationships are described using both "m" and "M" and a new entity is introduced to the diagram: student. Here, we assume that a specific class is always attended by more than one student. In addition, we assume that the probability of a specific class being taught by more than one teacher was determined to be below the domain-specific threshold. In other words, the final implication of the two assumptions regarding the treatment of the relationships would result as follows: the m:M relationship for teacher-class will be treated as 1:M, and class student relationship will be treated as standard M:M in the implementation.

To highlight the implications of the probabilistic cardinality notation, we discuss possible data modeling approaches to 1:m relationship and summarize them in Table 5 below. In particular, we underscore the implications for metric apportioning regarding each case. Specifically, when a relationship is designated as "M" or the relationship is broken into two relationships, whereby secondary relationship is designated "M", there will be a need for metric apportioning of a quantitative measure associated with instances of those relationships.

By enabling a more nuanced representation of relationships between dimensions and facts, probabilistic cardinality notation may enhance the mapping accuracy in the data warehousing context. This approach has the potential to reduce the risk of generating inaccurate metrics as analytical outputs can be more accurately grounded in the structure and semantics of the data and its relationships. The probabilistic cardinality notation is implemented during data modeling and has an impact on the subsequent stages of the big data analytics process.

Having identified the need for apportioning business analytics metrics, we now turn our attention to our proposed addition to current methodologies and approaches for apportioning metrics among multiple instances of dimension entities as they relate to a single instance of a fact entity.

4.2. Metric apportioning notation: replicability

This section introduces replicability, a metric apportioning notation. We present the semantics of this notation through dimensional modeling, framing our contribution in familiar terminology. Although the underlying issue extends to alternative modeling paradigms in the big data analytics framework, the data warehousing perspective offers a straightforward language for demonstration.

In organizational settings, attributes linked through many-to-many relationships frequently present potential metrics that capture aspects of the data represented in those relationships. The proposed metric apportioning notation extends standard data modeling semantics by clarifying how a metric is apportioned across the instances of its associated dimensions, thereby addressing a fundamental characteristic of business analytics metrics. Instances of metrics may correspond to instances of dimensions in different ways. The simplest and most straightforward common case is when a metric instance directly aligns with one particular instance of a related dimension. This is a typical case in data warehouses containing transactional data, but even in such settings this simple correspondence does not always hold. There are also cases where a single metric corresponds to multiple instances of a related dimension, and this can manifest in different ways. In some scenarios, the same metric value is replicated across all instances of a related dimension. In other scenarios, a single metric instance is related to two or more instances of a dimension, with its value apportioned in specific proportions among them. We introduce the notation of replicability to address the need for proper metric apportioning during the data modeling stage of big data analytics. Replicability distinguishes between two types of metrics, defined as follows:

Replicability is a metric apportioning notation that differentiates

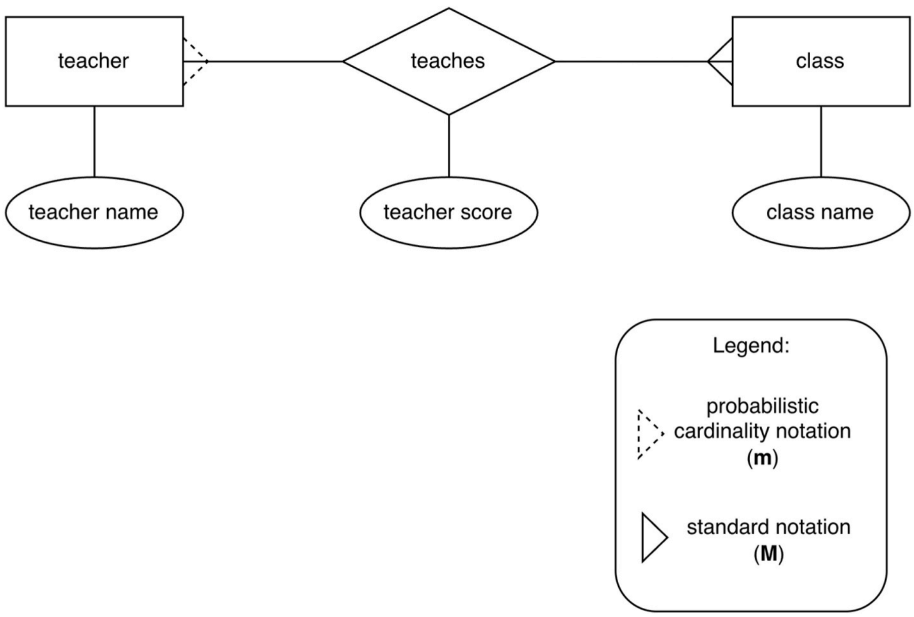


Fig. 2. ER diagram of teacher-class relationship with probabilistic cardinality notation.

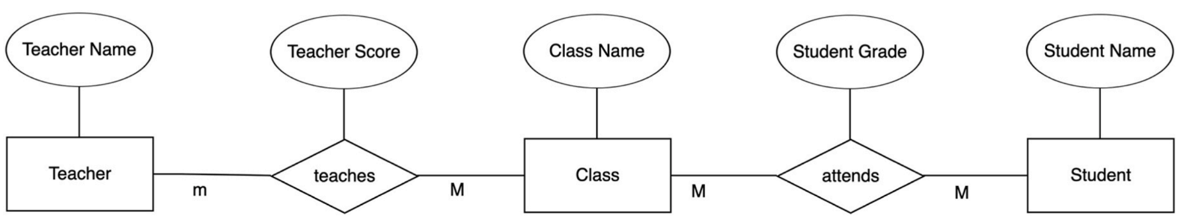


Fig. 3. ER diagram for teacher-class-student relationship using both "m" and "M" notation.

Table 5
Probabilistic cardinality notation in context of design strategies in big data analytics process.

How do we treat "m" in "1: m" relationship in the design stage?	Data Modeling Approach in Big Data Analytics Process	Need for metric apportioning?
Treat as "M".	Use standard approach for handling one-to-many relationships.	Yes.
Treat as "1".	Use standard approach for handling one-to-one relationships.	No.
Treat as two relationships: Primary as "1" and secondary as "M".	Use standard approach for one-to-one relationship for the primary relationship and standard one-to-many for the secondary relationship.	Yes.

between replicable and non-replicable metrics in the following way.

- **Non-replicable**: The metric is uniquely allocated with respect to a given dimension, such that its full value corresponds to only one instance of that dimension.
- **Replicable**: The metric is associated with multiple instances of a given dimension, with each instance sharing the full value of the metric.

Dimensional modeling distinguishes between two types of dimensions based on their relationship to the transaction entity. The first type consists of dimensions whose component entities are in a one-to-many relationship with the transaction entity. In such cases, the issue of metric apportioning does not exist because each instance of the fact is associated with one instance of a dimension. The second type consists of dimensions whose component entities are in a many-to-many relationship with the transaction entity. Here, the issue of metric apportioning becomes relevant. The applied cases in Section 5 illustrate how replicability notation can be implemented across different data capturing methods and data modeling approaches. In the context of dimensional modeling, the metrics may appear as facts in different fact tables. In the standard mapping, metrics are stored in the core fact table with all dimensions directly connected, which is suitable for non-replicable metrics where the apportioning ratio between dimension instances is known. Alternatively, metrics may be placed in a custom fact table that includes only the dimensions to which replicability applies. When all metrics are defined as replicable, the core fact table becomes a coverage fact table and metrics only appear in custom fact tables.

We provide a generic scenario to illustrate a scenario in big data analytics process where data sources contain a combination of replicable and non-replicable metrics. For replicable metrics, each value in a transaction entity that is in a many-to-many relationship with a classification entity is associated with multiple instances of a classification entity. For non-replicable metrics, each value of the metric is only associated with one instance of a classification entity, conditional on the participation condition attribute. The data sources are described in Table 6 in terms of the type of relation they contain and a brief description of each data source.

In Fig. 4, we present an ER diagram of the data model for this generic scenario and the table and column names that are later referred to in the code which are provided to contrast aggregation of replicable and non-replicable metrics.

Table 6
Data Sources for generic scenario containing both replicable and non-replicable metrics.

Data Source Name	Type	Description
Data Source 1	transaction entity relation	Contains replicable and non-replicable metrics.
Data Source 2	associative entity relation	Captures many to many relationship between transaction entity relation and classification entity relation. It contains a participation condition attribute which signifies whether an instance of Data Source 3 classifier appears in a relationship with Data Source 1 in a manner that a non-replicable metric applies to it.
Data Source 3	classification entity relation	Represents a relation that will be mapped into a dimension and contains a dimensional classifier by which the metrics will be grouped.

In addition, we provide relational algebra expressions for aggregations of replicable and non-replicable metrics in Table 7 below. When aggregating non-replicable metrics, the participation condition variable is considered only when it is true. In real scenarios, this condition depends on the context. For example, in a healthcare context, duration of the surgery would be counted only for the lead surgeon rather than adding up the times for every assisting staff member. In this way, the total surgery hours that took place in the hospital would be accurately represented. Another example is fleet operations in logistics, where fuel consumption for a trip may be attributed only to the driver operating the vehicle, but not all other employees associated with the trip record. Otherwise, the fuel consumption would be inaccurately multiplied across all employees, leading to inflated totals.

Finally, we illustrate the aggregation of replicable and non-replicable metrics through a general code example, situated within the analytics layer of the Big Data Analytics framework. The SQL implementation is shown in Table 8, where the distinction between the two cases is determined by the use of the participation condition in the WHERE clause.

This section has examined replicability at a granular level, framed within the data warehousing context, in order to introduce the concept through a generic scenario. The scenario reflects common situations where external data sources are integrated, and records of differing granularity describe the same event. Section 5, supported by an illustrative example, extends this discussion by demonstrating more concretely the implications of accounting for replicability in the data modeling stage of the big data analytics process.

5. Illustrative examples

In this section, we use illustrative examples of how probabilistic cardinality notation and replicability can be used in the context of our big data analytics framework. We provide two case studies and present the results from various stages of the big data analytics process below.

Table 7
Generic scenarios of aggregating non-replicable and replicable metrics using relational algebra.

Aggregation Metric Type	Relational Algebra Expressions for Metric Aggregation
Non-Replicable	γ_{Classifier_Name; SUM(Non-Replicable_Metric)} (σ_{DS2.Participation_Condition = TRUE} ((DS1 ⋈_{DS1.Primary_Key = DS2.Foreign_Key_DS1} DS2) ⋈_{DS2.Foreign_Key_DS3 = DS3.Primary_Key} DS3))
Replicable	γ_{Classifier_Name; SUM(Replicable_Metric)} ((DS1 ⋈_{DS1.Primary_Key = DS2.Foreign_Key_DS1} DS2) ⋈_{DS2.Foreign_Key_DS3 = DS3.Primary_Key} DS3)

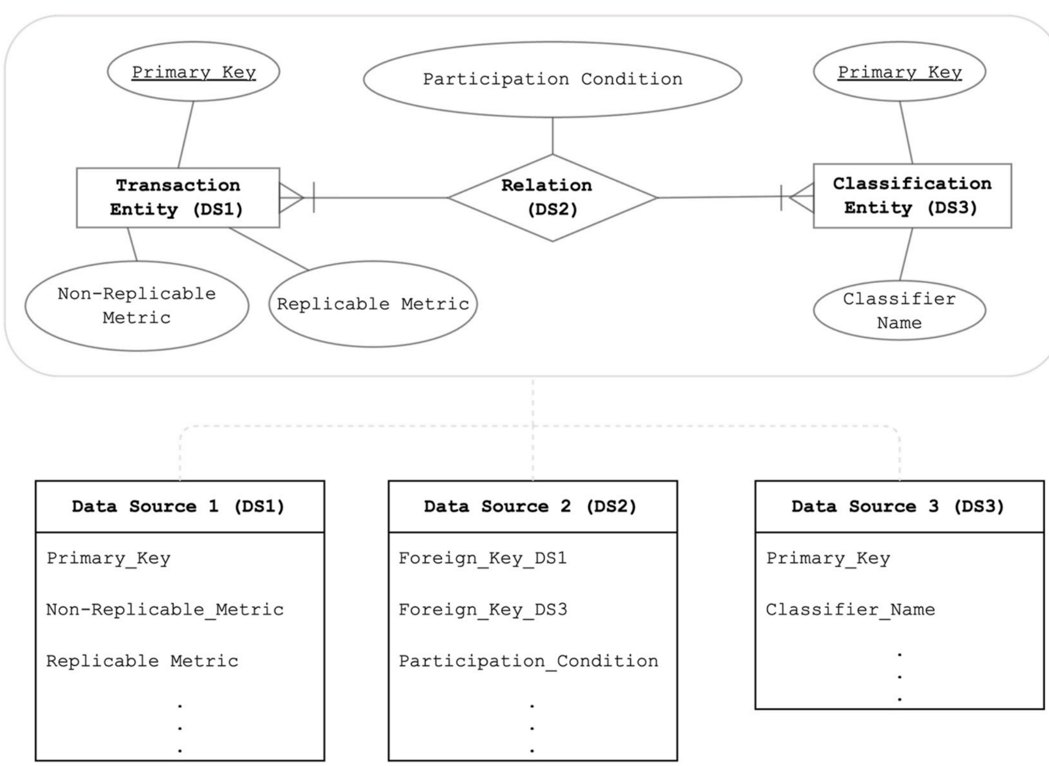


Fig. 4. Data model and data sources with replicable and non-replicable metrics.

Table 8
Generic scenarios of aggregating non-replicable and replicable metrics in SQL.

Aggregation Metric Type	SQL Code for Metric Aggregation
Non-Replicable	SELECT DS3.Classifier_Name, SUM(Non_Replicable_Metric) AS Metric_Value FROM Data_Source_1 DS1 JOIN Data_Source_2 DS2 ON DS1.Primary_Key = DS2.Foreign_Key_DS1 JOIN Data_Source_3 DS3 ON DS2.Foreign_Key_DS3 = DS3.Primary_Key WHERE DS2.Participation_Condition = TRUE GROUP BY DS3.Classifier_Name
Replicable	SELECT DS3.Classifier_Name, SUM(Replicable_Metric) AS Metric_Value FROM Data_Source_1 DS1 JOIN Data_Source_2 DS2 ON DS1.Primary_Key = DS2.Foreign_Key_DS1 JOIN Data_Source_3 DS3 ON DS2.Foreign_Key_DS3 = DS3.Primary_Key GROUP BY DS3.Classifier_Name

5.1. Case study 1: "EduMetrics" and probabilistic cardinality notation

This case study provides a use case of probabilistic cardinality notation in an organizational setting. For this purpose, we come up with a fictive consulting organization "EduMetrics" that leverages big data analytics where they use various metrics to benchmark school curricula and evaluate student satisfaction, thereby enabling data-driven insights for educational excellence. The big data analytics outputs developed by EduMetrics are utilized both by schools and government agencies in order to enhance educational outcomes and improve quality of the learning environment. The scope of their work is wide as they deliver solutions across municipal, county and state levels.

While EduMetrics develops many metrics for their clients, this section focuses on the specific metric: class evaluations. The purpose of this case study is to demonstrate how even with a simple relationship and just an excerpt of the data set, the resulting metrics based on the same event can differ based on the data modeling approach taken. The data sources used for the class evaluation metric in EduMetrics' big data analytics process vary across clients, as each organization records this metric differently. For example, some clients are able to use their technological infrastructure such as video class recordings and digital learning platform data to distinguish between main teacher, guest lecturer, substitute teacher, etc. This big data aspect means that the data modeling part of the process is more complex, even when it comes to simple relationships. To show how such a relationship can be impacted by introducing the probabilistic cardinality notation in the data modeling process, we focus on the Teacher-Class relationship for evaluation metrics.

For the purpose of contrasting the outcomes of various data modeling approaches and demonstrating how the same data source can be used across them, the unique data source includes records of four teachers teaching ten different classes: nine classes are taught by one teacher each, and one class is taught by three teachers. In addition, the following business-scope requirements were considered.

- each teacher must teach at least two classes,
- all teachers routinely teach multiple classes,
- most of the classes are taught by a single teacher, although the data shows that a small proportion of classes are taught by more than one teacher.

In the context of this case study, teaching is used as a broad term to describe any instructional participation in a class, including serving as a full-time instructor for a semester, acting as a guest lecturer, or substituting for another teacher. Under standard ER modeling notation, the Teacher-Class relationship is represented as a conventional many-to-many relationship. In contrast, we present several data models that can be derived through the application of probabilistic cardinality notation. For each data model, we provide a corresponding ER diagram as well as populated data tables.

Fig. 5 depicts the ER diagram for the Teacher-Class relationship with its associated attributes, applying probabilistic cardinality notation "m" and summarizing the approaches to treating "m". For each approach we provide the resulting data models and corresponding populated data tables for both relational and non-relational implementations.

We assume that EduMetrics can rely on different approaches to determining how the probabilistic cardinality m is treated in the Teacher–Class relationship. In a process-driven procedure, organizational stakeholders decide how multi-teacher situations should be interpreted based on evaluation policy rather than observed frequencies. In this approach, the treatment of m encodes these policy decisions directly. For example, if the organization intends to treat each class as having a single responsible teacher, it assigns the value $m(1)$. If all teachers are considered equally responsible and evaluated individually, the appropriate designation is $m(M)$. If the organizational goal is to distinguish between contributions of a primary teacher and additional teachers, they select the case $m(1,M)$. In this way, the process-driven approach directly determines the treatment of m in the Teacher–Class relationship.

In contrast, data-driven calibration of the threshold τ is based on repeated empirical evaluation of EduMetrics' historical Teacher–Class data. In this approach, the probability $p(e_1)$, the likelihood that a class is taught by more than one instructor, is estimated from historical assignments and typically remains stable for a given dataset (e.g., $p(e_1) = 0.10$). The threshold τ, however, may be re-estimated periodically to reflect changes in instructional patterns or domain drift across semesters or client contexts. When τ is recalibrated in this way, different threshold values (e.g., 0.15, 0.10, or 0.05) lead to different interpretations of the probabilistic cardinality even though the underlying probability remains fixed. As shown in Table 9, a threshold of $\tau = 0.15$ results in $m(1)$, a threshold of $\tau = 0.10$ yields the mixed case $m(1,M)$, and a threshold of $\tau = 0.05$ produces $m(M)$. This illustrates that data-driven calibration allows EduMetrics to adopt different treatments of the Teacher–Class relationship solely by re-estimating the threshold while remaining based on the same historical data.

Data-driven and process-driven determinations of m do not operate in isolation. The observed probabilities and the threshold values ultimately lead to uncovering the organization's underlying business rules, and when those rules evolve, both the empirical probabilities and the corresponding threshold will change as well. Thus, while the two approaches differ methodologically, they remain interconnected and mutually reinforced in practice.

Each of the three approaches has the same information available from their data sets and when presenting the conceptual data models and the resulting tables, we highlight the row(s) where Class_ID = 10 shows up as this class is the one where there is information on multiple instructors teaching the course and it allows for more granular level of information for the evaluation score metric.

The first relational approach (R1) is the one where "m" is treated as 1. In this case, only the primary teacher is relevant for the business case. The evaluation score is in this case an attribute of the class, and the relationship is presented in Fig. 6. The resulting tables are Teacher and Class, where each row in the Class table represents unique instance of Class_ID with class name, evaluation score and Teacher_ID.

Another approach is to treat "m" as a standard M and map the relationship as relational as shown in Fig. 7. This approach results in three tables: Teacher, teaches and Class. Each row of the "Teaches" table consists of a unique combination of Teacher_ID and Class_ID, where each unique combination has its respective primary designation flag to designate if the teacher is the primary course instructor as well as the Evaluation Score.

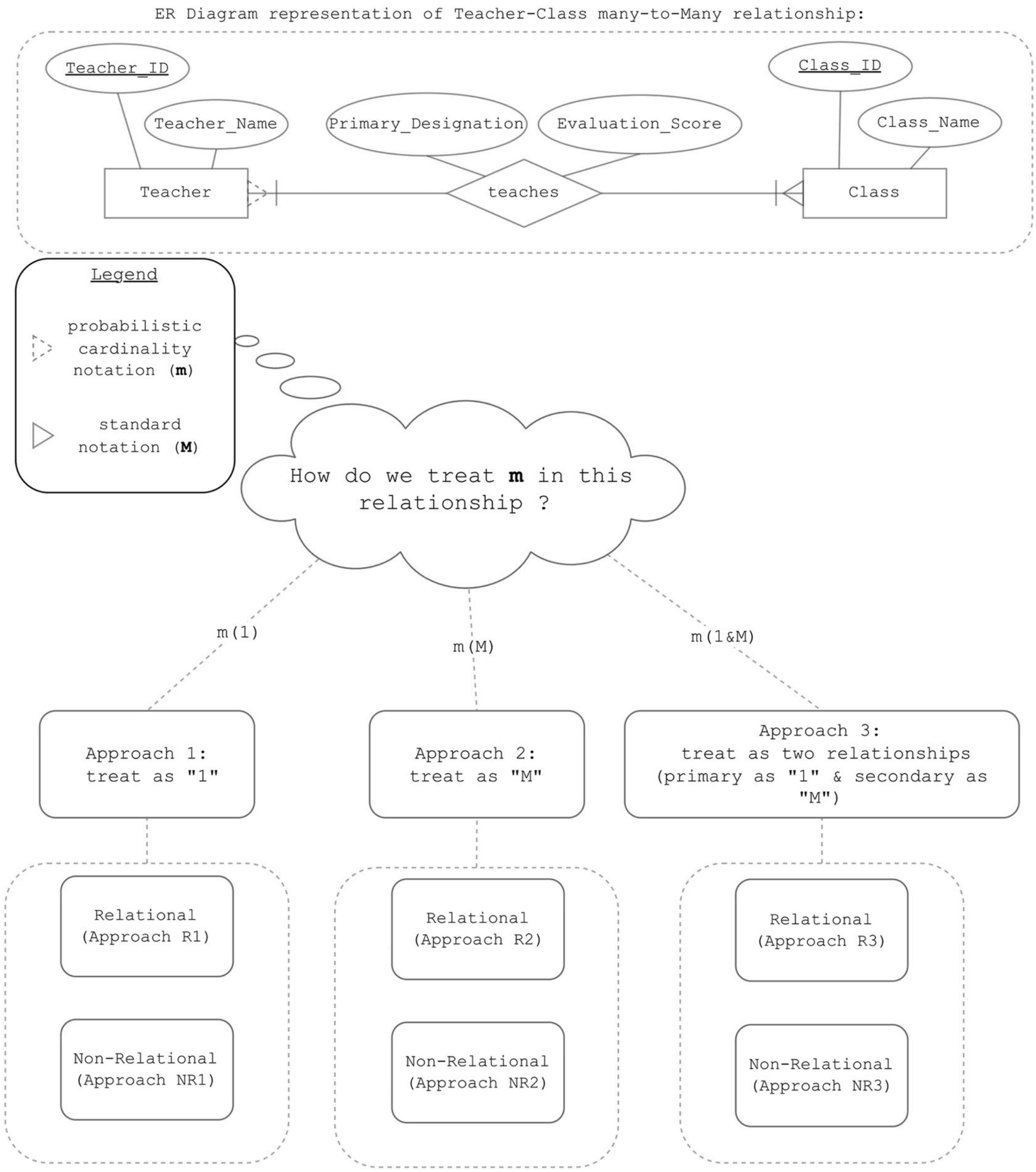


Fig. 5. Approaches for modeling "many-to-many" relationship for teacher-class relationship.

Table 9
Data-driven calibration via re-estimation of τ

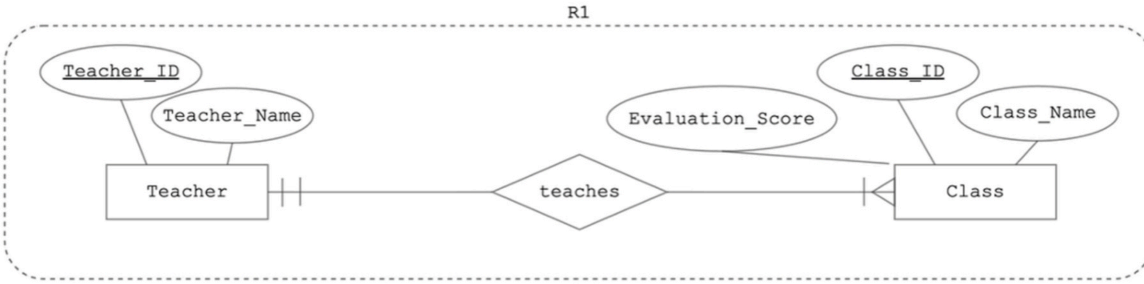
Treatment of m	$p(e_1)$	τ	Comparison	Interpretation
m(1)	0.1	0.15	(0.1 < 0.15)	Treat Teacher–Class as 1:1 (primary teacher only)
m(M)	0.1	0.05	(0.1 > 0.05)	Treat as 1:M (all teachers evaluated separately)
m(1,M)	0.1	0.10	(0.1 = 0.10)	Decompose into 1:1 + 1:M (primary + secondary instructors)

The third relational approach, presented in Fig. 8, is where "m" is treated as two relationships: primary as "1", and secondary relationship as "M". In this case there are two relationships between teacher and class. Here, the primary teacher gets primary evaluation score designated and tracked through the primary relationship and secondary teacher(s) have the evaluation score recorded as "Evaluation_Score" attribute of "teaches secondary" relationship.

Fig. 9 represents a non-relational approach NR1, where teacher performance is a single-valued composite attribute of class entity. In this approach, teacher performance is an attribute of class and evaluation score is attribute of the teacher performance. The columns for each row of the resulting Class table consist of: Class Name, Teacher ID, Teacher Name, and Evaluation Score.

Non-relational approach (or NoSQL type of approach) NR2 is presented in Fig. 10. Here, the "teacher performance" is a multivalued composite attribute of entity "class" and evaluation score is attribute of the "teacher performance". The resulting table is "Class" table with unique Class_ID for each record and columns class name and "Teacher Performance". In this case, the "Teacher Performance" column is in JSON format.

Finally, Fig. 11 depicts a non-relational approach as a combination of the scenarios where class has two attributes: Primary Teacher Performance and Secondary Teacher Performance. The Secondary Teacher Performance attribute is optional and multivalued and both Teacher Performance attributes have individual Evaluation_Score, Teacher_ID and Teacher_Name attributes. The resulting table has unique column for Class_ID and other columns are Primary Teacher Name and Primary Evaluation Score as well as sparsely populated Secondary Teacher



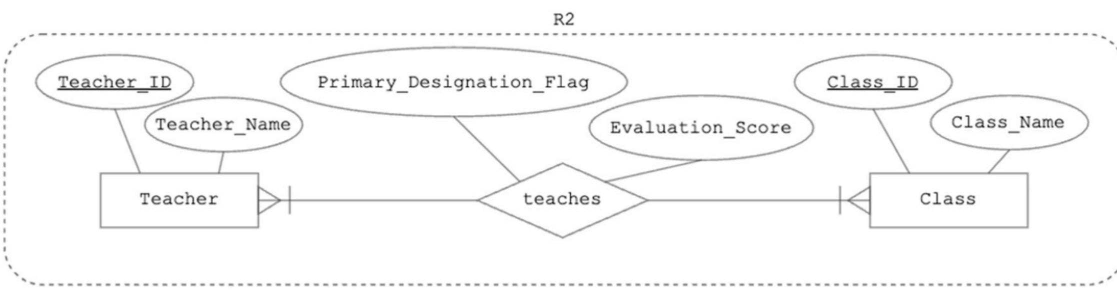
Teacher Table

Teacher_ID	Primary_Teacher_Name
1	Alice Johnson
2	Bob Smith
3	Carol Lee
4	David Kim

Class Table

Class_ID	Class_Name	Evaluation_Score	Teacher_ID
1	Math 101	4,06	1
2	Physics 101	3,34	2
3	Chemistry 101	4,97	3
4	Biology 101	3,83	4
5	English 101	4,18	1
6	History 101	4,54	2
7	Art 101	4,30	3
8	Music 101	3,31	4
9	PE 101	3,65	1
10	CompSci 101	4,58	2

Fig. 6. Relational approach where “m” is treated as “1” (approach R1).



Teacher Table

Teacher_ID	Teacher_Name
1	Alice Johnson
2	Bob Smith
3	Carol Lee
4	David Kim

teaches Table

Teacher_ID	Class_ID	Primary_Designation_Flag	Evaluation_Score
1	1	1	4,06
2	2	1	3,34
3	3	1	4,97
4	4	1	3,83
1	5	1	4,18
2	6	1	4,54
3	7	1	4,30
4	8	1	3,31
1	9	1	3,65
2	10	1	4,58
3	10	0	3,04
4	10	0	3,28

Class Table

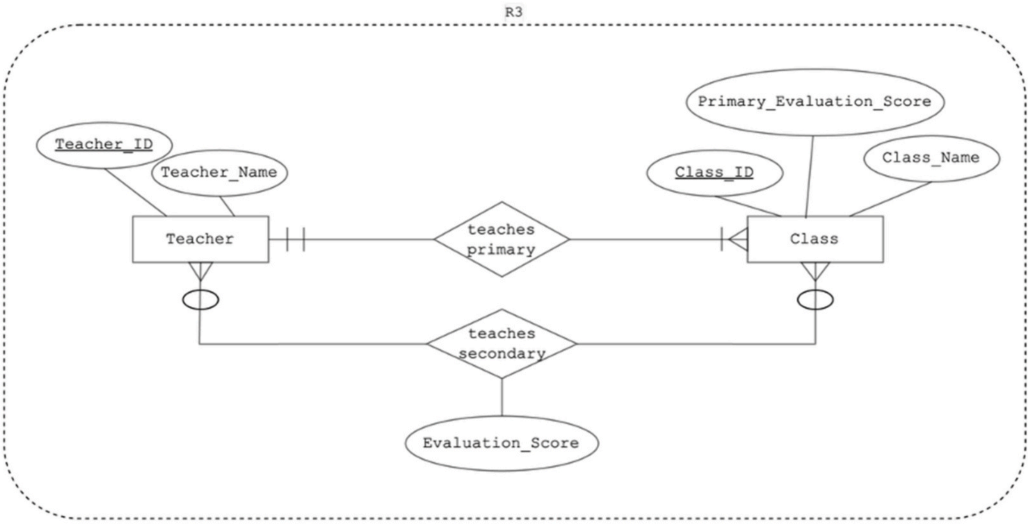
Class_ID	Class_Name
1	Math 101
2	Physics 101
3	Chemistry 101
4	Biology 101
5	English 101
6	History 101
7	Art 101
8	Music 101
9	PE 101
10	CompSci 101

Fig. 7. Relational approach where “m” is treated as “M” (approach R2).

Performance Column.

It is important to note that all of these approaches are correct, as long as the data model fits the requirements and needs of the big data analytics process. The purpose of demonstrating all six approaches is not to rank or declare one is better than another but to depict how even in a small context of two entities and data providing information on only 10 instances of the event, the data modeling approach can affect simple metrics. Again, the data is stored in appropriate tables for each approach and the metrics we summarize in Tables 10 and 11 below are all correct with respect to the approach taken. However, there is currently no formal way to recognize or address which way to go in the data modeling stage and the decision on how to model is up to the stakeholder performing analysis. Furthermore, different analysts might come up with different understanding of the business context and implicitly do that through their code without ever creating the conceptual data model.

Table 10 below shows the comparison of averaging the metric of interest as influenced by the approach taken in treating the “m” relationship. Relational and non-relational Approach 1 differ from Approaches 2 and 3, as further illustrated in Table 11, where each row lists the average values for each class within an approach. For Class_ID = 10, the averages are higher under Approach 1 than under the other two approaches, which increases the overall average evaluation score for Approach 1 (R1 & NR1) in Table 10. This difference arises from the way the metric is apportioned across dimension instances, highlighting how replicability directly affects aggregation results. Even in a small example with limited data, the choice of modeling approach can alter seemingly simple metrics, underscoring the need to explicitly account for



Class Table

Class_ID	Class_Name	Primary_Teacher_ID	Primary_Evaluation_Score
1	Math 101	1	4,06
2	Physics 101	2	3,34
3	Chemistry 101	3	4,97
4	Biology 101	4	3,83
5	English 101	1	4,18
6	History 101	2	4,54
7	Art 101	3	4,30
8	Music 101	4	3,31
9	PE 101	1	3,65
10	CompSci 101	2	4,58

Teacher Table

Teacher_ID	Teacher_Name
1	Alice Johnson
2	Bob Smith
3	Carol Lee
4	David Kim

teaches secondary Table

Teacher_ID	Class_ID	Evaluation_Score
3	10	3,04
4	10	3,28

Fig. 8. Relational Approach where “m” is treated as “1” and secondary relationship of “m” (approach R3).

replicability during the data modeling stage.

Finally, we provide basic SQL queries that retrieve the class name and its evaluation score, allowing us to contrast the queries for relational approaches R1, R2, and R3. These queries demonstrate the variety of ways analysts may calculate the evaluation score of a class. The SQL is kept consistent with the approaches outlined above, ensuring that the differences observed in the models are also reflected directly in the code. If the business case scope is considering the fact that they have one evaluation for one teacher for one class, the following SQL query is used as the organization treats “m” as “1” (approach R1):

```
SELECT Class_Name, Teacher_Score

FROM Class
```

If the “m” is treated as “M” (approach R2), the following SQL query is used:

```
SELECT c.Class_Name, AVG(t.Teacher_Score)

FROM Class c

JOIN teaches t ON c.Class_ID = t.Class_ID

GROUP BY c.Class_Name
```

If the “m” is treated as two relationships (Approach R3), the stakeholder can decide whether to keep the metrics separate—primary class evaluation and secondary class evaluation scores—or to combine them into a single metric. While we provide the SQL code reflecting the combined approach in R3, the separate treatment can also be implemented. For example, the following query retrieves the Primary Class Evaluation:

```
SELECT Class_Name, Primary_Evaluation_Score

FROM Class
```

For the secondary class evaluation score the SQL query is as follows:

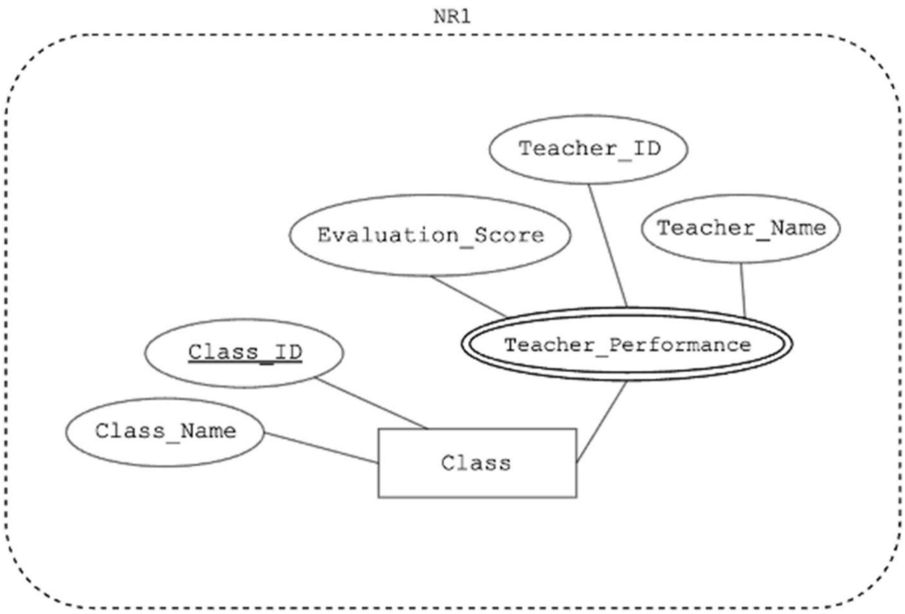
```
SELECT Class_Name, AVG(Evaluation_Score)

FROM Class c

JOIN teaches_secondary ts ON c.Class_ID = ts.Class_ID

GROUP BY Class_Name
```

For the option of combining the metrics into one, a straightforward approach is to calculate a regular average of both the primary and secondary evaluation scores. This method treats both sources equally, without assigning different proportions. This was the approach taken in R3, and the SQL query below shows how it can be implemented:

```
SELECT
C.Class_Name,
AVG(
CASE.
WHEN ts.Evaluation_Score IS NOT NULL.
THEN ts.Evaluation_Score.
ELSE c.Primary_Evaluation_Score.
END.
) AS Evaluation_Score.
FROM.
Class c.
LEFT JOIN
Teaches_secondary ts.
ON c.Class_ID = ts.Class_ID.
GROUP BY
C.Class_Name; c.Class_Name.
```

5.2. Case study 2: “VroomRide” and replicability notation

This case study depicts a fictive ridesharing organization “VroomRide” that keeps track of data on ride transactions involving various passengers, vehicles and drivers. The metrics that the organization is



Class Table

Class_ID	Class_Name	Teacher_ID	Teacher_Name	Evaluation_Score
1	Math 101	1	Alice Johnson	4,06
2	Physics 101	2	Bob Smith	3,34
3	Chemistry 101	3	Carol Lee	4,97
4	Biology 101	4	David Kim	3,83
5	English 101	1	Alice Johnson	4,18
6	History 101	2	Bob Smith	4,54
7	Art 101	3	Carol Lee	4,30
8	Music 101	4	David Kim	3,31
9	PE 101	1	Alice Johnson	3,65
10	CompSci 101	2	Bob Smith	4,58

Fig. 9. Non-Relational Approach where “m” is treated as “1” (approach NR1).

keeping track of that are associated with each ride transaction are miles driven, dollars charged, rating and waiting time. In order to contrast the data modeling outcomes when replicability is introduced, we discuss various scenarios that represent different data capturing methods and data modeling approaches. All of the scenarios share the following business scope assumptions.

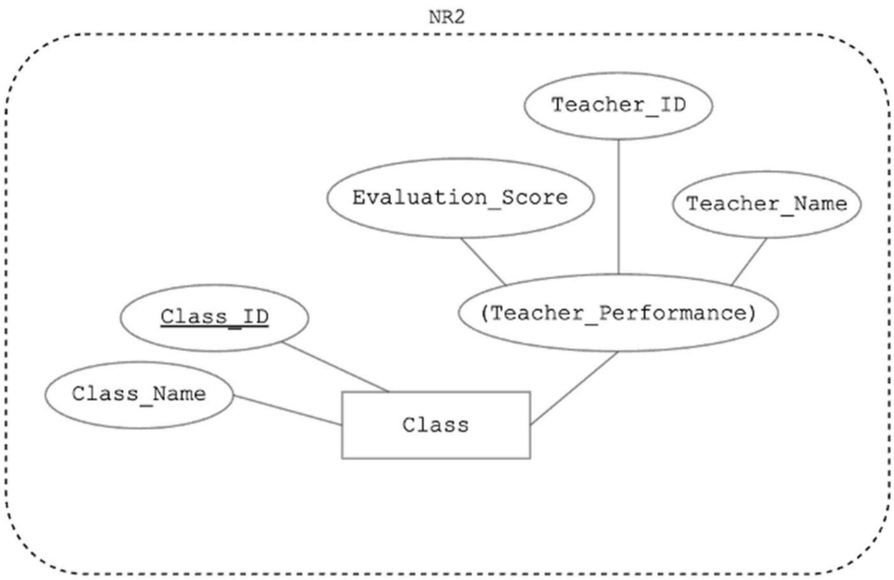
- each ride happens on one calendar day (even if it starts before midnight and finishes after midnight) with all metrics allocated to the day the ride had started,
- each ride has only one driver,
- each ride can have one or more passengers.

In this case study, we continue to use a relational modeling/data warehousing approach, and provide ER diagrams, relational schemas and dimensional models as well as snapshots from the Tableau analytics dashboard to show implications of recognizing replicability in the modeling process on the consistent metric reporting in big data analytics.

Table 12 provides a summary of all of the scenarios included in this case study. The scenarios are numbered and named based on their key differentiating features: Data Sources, Data Modeling Approaches, and recognition of replicability in the big data analytics process. In the Big Data Relational and Big Data Dimensional Approaches the data modeling process recognizes and accounts for replicability, while the Old Paradigm Dimensional Approach is the traditional approach to data modeling that does not formally recognize replicability. The baseline scenario is used as a benchmark case for the entire case study.

5.2.1. *Case study 2: baseline scenario*

In the baseline scenario, there may be more than one passenger participating in a ride transaction, but only one passenger is recorded in transactional database. Each passenger that is recorded for the specific ride transaction is the one that is paying for the transaction and as such is considered to be the primary passenger. Here, “VroomRide” does not keep track of anyone else riding along with the primary passenger and all the data generated from enterprise application software is going into the transactional databases and makes for a single data source in this scenario. In other words, the key differentiating feature between baseline scenario and other scenarios is that there is no access to external big data information sources and the issue of apportioning the metrics does not exist. The metrics collected for each individual ride transaction are Miles Driven, Dollars Charged, Rating and Waiting Time and the data source consists of five unique drivers, three passengers that are directly involved in payment transactions, 20 instances of ride transactions and six different vehicles (for more details and all the data, see supplemental code in the repository linked in the appendix). The ride transaction records contain data for every ride transaction and the participating passenger, driver and vehicle as well as the metrics. ER model, relational model and dimensional model for this scenario are represented in Table 13 below. The ER model has 4 entities: passenger, ride transaction, vehicle and driver. The relational model has ride transaction table with



Class Table

Class_ID	Class_Name	Teacher_Performance
1	Math 101	[{"Teacher_ID": 1, "Teacher_Name": "Alice Johnson", "Evaluation_Score": 3.12}]
2	Physics 101	[{"Teacher_ID": 2, "Teacher_Name": "Bob Smith", "Evaluation_Score": 4.83}]
3	Chemistry 101	[{"Teacher_ID": 3, "Teacher_Name": "Carol Lee", "Evaluation_Score": 3.65}]
4	Biology 101	[{"Teacher_ID": 4, "Teacher_Name": "David Kim", "Evaluation_Score": 2.89}]
5	English 101	[{"Teacher_ID": 1, "Teacher_Name": "Alice Johnson", "Evaluation_Score": 2.53}]
6	History 101	[{"Teacher_ID": 2, "Teacher_Name": "Bob Smith", "Evaluation_Score": 3.28}]
7	Art 101	[{"Teacher_ID": 3, "Teacher_Name": "Carol Lee", "Evaluation_Score": 4.04}]
8	Music 101	[{"Teacher_ID": 4, "Teacher_Name": "David Kim", "Evaluation_Score": 5.0}]
9	PE 101	[{"Teacher_ID": 1, "Teacher_Name": "Alice Johnson", "Evaluation_Score": 4.49}]
10	CompSci 101	[{"Teacher_ID": 2, "Teacher_Name": "Bob Smith", "Evaluation_Score": 4.12}, {"Teacher_ID": 3, "Teacher_Name": "Carol Lee", "Evaluation_Score": 4.22}, {"Teacher_ID": 4, "Teacher_Name": "David Kim", "Evaluation_Score": 2.81}]

Fig. 10. Non-relational approach where “m” is treated as “M” (approach NR2).

Transaction ID as primary key where foreign keys are Passenger ID, Vehicle VIN and Driver ID coming from Passenger, Vehicle and Driver tables. Finally, the dimensional model consists of one core fact table connected with vehicle, passenger, driver and calendar dimensions.

Both dimensional and relational modeling approaches are simple in terms of metric apportioning as each record of Ride_Transaction and Core fact tables contains data on one participating passenger and driver. In other words, metrics are apportioned with respect to one driver and one passenger for one ride.

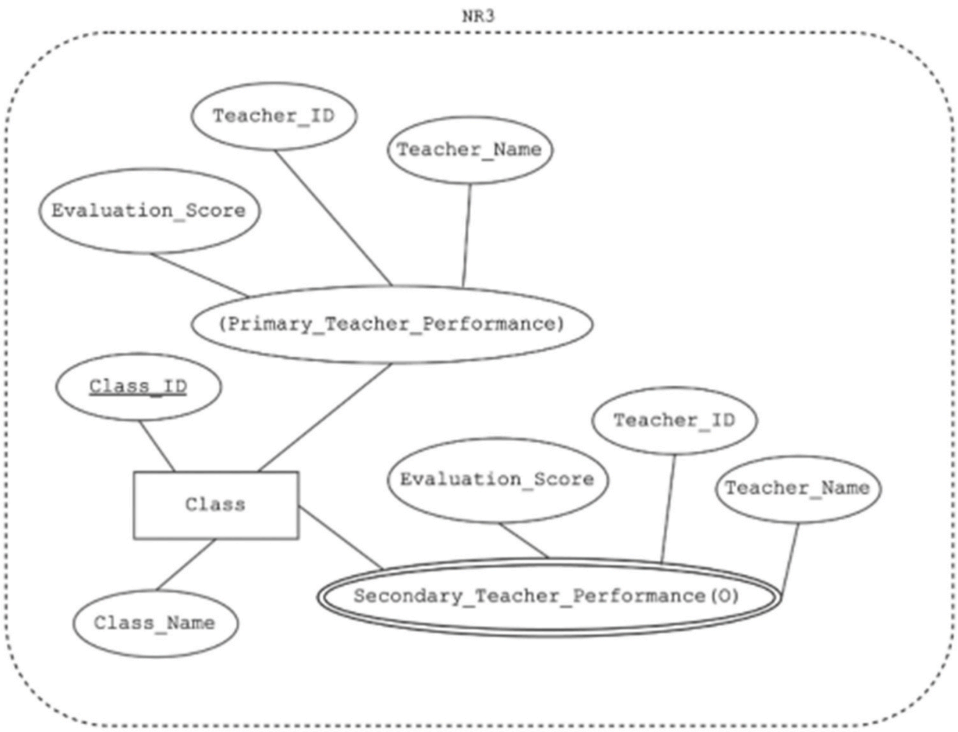
Business requirement from different business units of “VroomRide” was to create a dashboard with following metric totals: Passenger Mileage, Passenger Revenue, Driver Mileage, Driver Revenue and Driver Waiting Time. In Fig. 12, we provide the Tableau Dashboard the metrics listed as the business requirement both for relational and dimensional approaches.

5.2.2. *Case study 2: big data scenarios*

In the following scenarios, “VroomRide” is able to keep track of additional passengers for each transaction through their big data source. Data sources for the big data analytics process now include transactional data coming from enterprise software application and big data where the location data is used to identify additional passengers for each ride transaction. In other words, the big data source means that the “VroomRide” now has data for five unique drivers, 20 unique passengers out of which three are directly involved in payment transactions and designated as “primary passengers”, 20 instances of ride transactions and six different vehicles (see repository linked in the appendix for the actual data).

We demonstrate different data modeling approaches the organization can take and how recognition of replicability in data modeling process reflects on accuracy of metric reporting in big data analytics. Here, we assume that the organizational stakeholders involved in the process such as data architects, software engineers or business analysts inherently understand the data model and structure and therefore enforce replicability through consistent application of that understanding. This assumption is revisited and critiqued in the discussion section.

First, we provide data models for the relational approach scenario in Table 14. Here, “Primary Passenger” is added as an attribute to the relationship between Passenger and Ride Transaction to distinguish between primary paying passenger and others. Furthermore, this model assumes that replicability is recognized in the data modeling process and denoted in the data models. The attributes “Miles Driven” and “Waiting Time” are denoted as replicable as more passengers share the same value of those metrics. The “(R)” next to the relationship denotes that the relationship between passenger and ride transaction is the one that the metrics are replicable with respect to. This also implies that dollars charged, and rating are non-replicable meaning only one instance of a



Class Table

Class_ID	Class_Name	Primary_Teacher_Name	Primary_Evaluation_Score	Secondary_Teacher_Performance
1	Math 101	Alice Johnson	4,06	[]
2	Physics 101	Bob Smith	3,34	[]
3	Chemistry 101	Carol Lee	4,97	[]
4	Biology 101	David Kim	3,83	[]
5	English 101	Alice Johnson	4,18	[]
6	History 101	Bob Smith	4,54	[]
7	Art 101	Carol Lee	4,30	[]
8	Music 101	David Kim	3,31	[]
9	PE 101	Alice Johnson	3,65	[]
10	CompSci 101	Bob Smith	4,58	[{"Secondary_Teacher_ID": 3, "Secondary_Teacher_Name": "Carol Lee", "Secondary_Evaluation_Score": 3.04}, {"Secondary_Teacher_ID": 4, "Secondary_Teacher_Name": "David Kim", "Secondary_Evaluation_Score": 4.28}]

Fig. 11. Non-relational Approach where "m" is treated as "1" and a secondary relationship of "m" (approach NR3).

Table 10
Overall Average Evaluation Score for different approaches to treating "m".

Approach	Overall_Average_Evaluation_Score
NR1	4,08
NR2	3,98
NR3	3,98
R1	4,08
R2	3,98
R3	3,98

Table 11
Evaluation Score for each class and each approach.

Class_ID	NR1	NR2	NR3	R1	R2	R3
1	4,06	4,06	4,06	4,06	4,06	4,06
2	3,34	3,34	3,34	3,34	3,34	3,34
3	4,97	4,97	4,97	4,97	4,97	4,97
4	3,83	3,83	3,83	3,83	3,83	3,83
5	4,18	4,18	4,18	4,18	4,18	4,18
6	4,54	4,54	4,54	4,54	4,54	4,54
7	4,30	4,30	4,30	4,30	4,30	4,30
8	3,31	3,31	3,31	3,31	3,31	3,31
9	3,65	3,65	3,65	3,65	3,65	3,65
10	4,58	3,63	3,63	4,58	3,63	3,63

passenger is associated with the full value of each instance of the metric.

In Table 15, we provide dimensional models (star schemas) for dimensional approaches that use big data as a data source on top of transactional data. These dimensional models are products of the data warehouse design process with the key difference between the two being one formally acknowledging replicability in the data modeling process while the other labeled "Old Paradigm" does not. The Big Data Dimensional Modeling approach consists of the ER diagram and the

Table 12
Case study 2 scenarios summary.

Scenario Number/ Scenario Name	Data Sources	Data Modeling Approaches	Replicability?	Accurate metric reporting?
0/Baseline Scenario	Relational database	ER, Relational, Dimensional	N/A (Replicability does not appear.)	Yes.
1/Big Data Relational Approach	Relational database, Big Data	ER, Relational	Replicability recognized and accounted for.	Conditional yes.
2/Old Paradigm Dimensional Approach	Relational database, Big Data	ER, Relational, Dimensional	Replicability not recognized.	No.
3/Big Data Dimensional Approach	Relational database, Big Data	ER, Relational, Dimensional	Replicability recognized and accounted for.	Conditional yes.

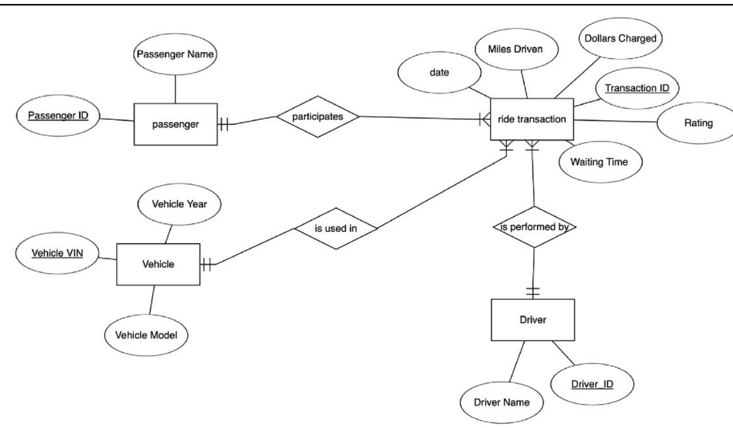
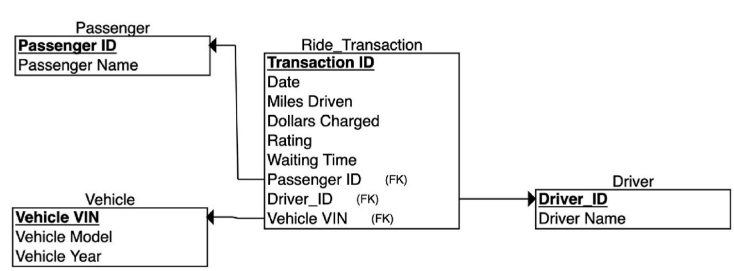
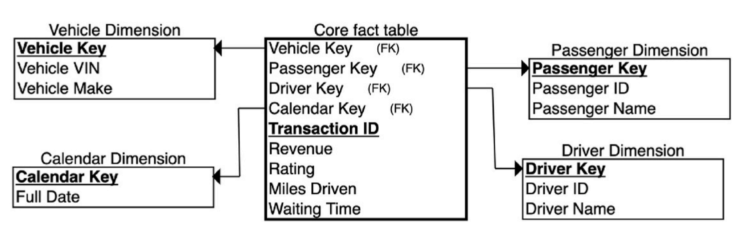
relational model presented in Table 14 and the Old Paradigm Dimensional Modeling approach ER model and relational schema can be found in Table A3. Dimensional Model for Scenario 2 has one core fact table that has a composite primary key consisting of Passenger Key and Transaction ID. Scenario 3 has three fact tables as a result of replicability being acknowledged. First is the "Revenue and rating fact table" where the only instances of passenger key that appear in this table are the ones that correspond to the instances of passengers who are primary passengers. Second one is the "Passenger, Mileage and Waiting Time fact table" where single dimension aggregations are only allowed for dimensions whose key value comes with label "R" and for the Calendar Dimension (when aggregation pertains to the dimension with the label "R"). Other dimensions can be used in aggregations but only in conjunction with dimensions labeled with "R", never on their own. Third table is "Driver, Mileage and Waiting time fact table" as miles driven and waiting time are non-replicable with respect to the driver dimension.

The Tableau dashboard, presented in Fig. 13, was created by connecting SQL databases for Scenarios 1, 2 and 3. Business requirement from different business units of VroomRide was to have the aggregated value of the following metrics included in the dashboard: Passenger Mileage, Driver Mileage, Passenger Revenue, Driver Revenue and Driver Waiting Time. The metrics that were aggregated incorrectly are Driver Mileage and Driver Waiting Time for Scenario 2.

Finally, to provide perspective on the scale of our database and its implications for analysis, Table 16 reports the number of relational tables containing the source data, the number of dimensions, the number of fact tables, the total number of two-way metric aggregations considered, the total number of two-way incorrect aggregations, and the resulting aggregation computation accuracy (%).

The difference in the number of fact tables can be explained by the use of replicability notation in the conceptual data modeling stage of the Big Data scenario, which influenced the relational database design and ultimately resulted in the data warehousing process generating more tables, as the metrics had to be grouped separately rather than in a single core fact table. Finally, the same five aggregations were calculated, but

Table 13
Data models for baseline scenario for case study 2.

Data Modeling Approach	Data Model
ER Model	
Relational Model	
Dimensional Model	

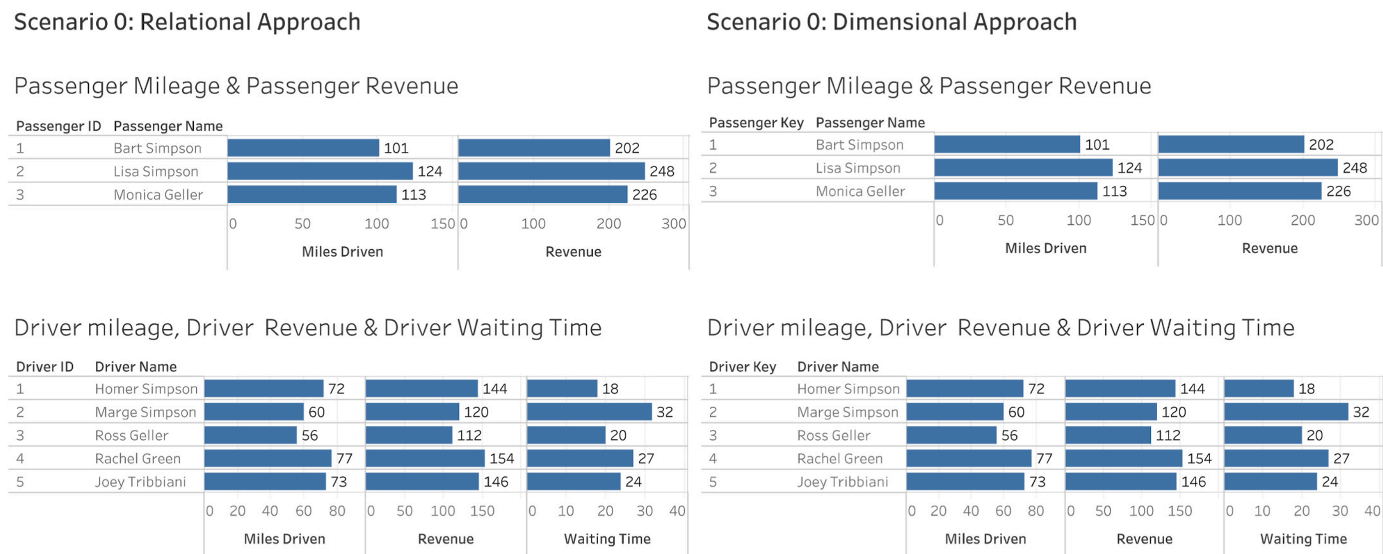
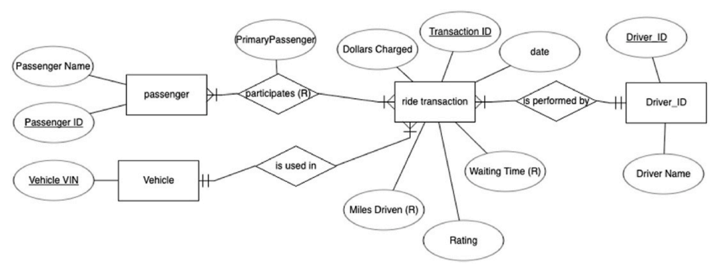
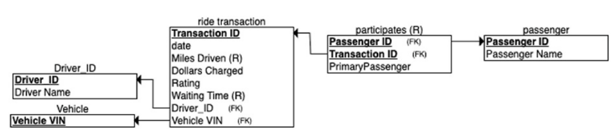


Fig. 12. Tableau dashboard for baseline scenario.

Table 14
Data models for scenario 1: Big data relational approach.

Data Modeling Approach	Data Model
ER Model	
Relational Model	

in the Old Paradigm two of them produced inaccurate results. In simple terms, this means that the Old Paradigm achieved 60 % accuracy, while the Big Data approach that considered replicability achieved 100 %. An important note is that these accuracy numbers assume the rest of the process was completely accurate; otherwise, the figures would be lower for both approaches.

6. Discussion

In this section, we discuss the theoretical and practical contributions of our proposed big data analytics framework and semantic data modeling additions for big data analytics.

6.1. Big data analytics framework

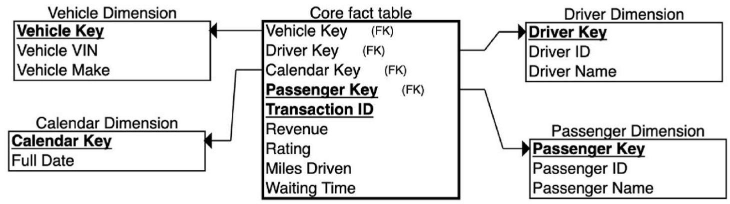
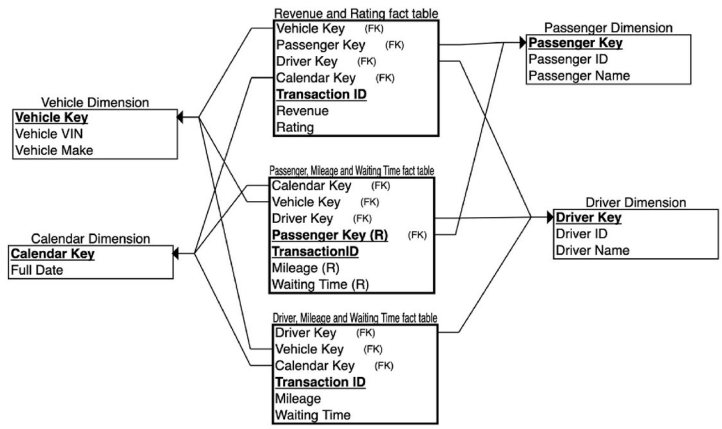
While the framework was created to give broader context to semantic data modeling additions, it is a contribution on its own. Theoretically, this framework is structured yet flexible, clearly outlining key components of the big data analytics process and accommodates diverse practices and contextual variations. Because of its conceptual simplicity, the framework is not restrictive; it acknowledges that each stage can be executed in a variety of ways, depending on the organizational context and strategic priorities.

Numerous big data analytics frameworks have been proposed in the literature, encompassing both process-oriented perspectives and classifications of key technical and methodological components of big data architectures. Although these frameworks include elements comparable to those in our proposed framework—such as data processing, data formats, ETL, visualization, preprocessing and data cleaning, and data analytics [65,72,73], none explicitly feature the data model as a central component. In contrast, our framework emphasizes the data model as a critical element whose quality and accuracy fundamentally influence all subsequent stages and components of the big data analytics process.

Although our primary focus is on data modeling practices addressing the challenges of integrating big data with other data sources, this framework offers a structured yet adaptable foundation that other researchers can leverage to explore various aspects of big data analytics processes. The framework supports various data modeling approaches, incorporating the use of both schema-on-read and schema-on-write implementations within the analytics value chain. It can serve as a tool to foster common documentation approaches and reference models in data governance, fostering shared process understanding and assisting organizations in formalizing and customizing their specific implementations.

Depending on the software systems that complement the analytics process, the framework could also be extended with system-level

Table 15
Data models for scenarios with dimensional approaches.

Scenario Number/Name	Dimensional Model
2/Old Paradigm Dimensional Approach	
3/Big Data Dimensional Approach	

information to support practical implementation. In this way, organizations and stakeholders can determine how prototype plug-ins or extensions would work best within their existing data modeling tools to represent replicability and probabilistic cardinality notations. Such adaptability allows the framework to be aligned with specific technological environments and organizational practices, further bridging the conceptual and applied aspects of the proposed approach.

6.2. Probabilistic cardinality notation

From a theoretical perspective, the probabilistic cardinality notation builds upon established data modeling practices and extends them to address a recurring problem in big data analytics that can arise when integrating heterogeneous data sources. Despite having its origins in modeling standard operational databases, since 1970s, ER modeling remains the most commonly used approach at the conceptual abstraction level in big data modeling [13]. Therefore, our concepts and illustrative examples are presented through that approach. As big data has the potential to change many relationship's cardinalities in an organization data model, we propose a declarative and high-level ER model extension to be integrated in the data modeling phase which is consistent with Badia's recommendation for ER model enhancements [46]. Although conceptual models have traditionally been used by IT experts for the purpose of designing databases, the use of conceptual models can also be beneficial in engaging non-technical stakeholders [43,44]. Therefore, an ER model enhanced with probabilistic cardinality notation has the potential to be useful to both technical and non-technical stakeholders in the big data analytics process by ensuring proper design based on a common understanding of business requirements.

This notation formally acknowledges the nuances of relationships in the data modeling process to eliminate the need for data modelers to make implicit and undocumented assumptions about the true nature of the relationships between various entities. Such practice could lead to less than optimal design that does not fully correspond to business requirements, leading to implementation that causes inconsistent metric reporting. Furthermore, in the absence of a clear and accurate data model, analysts usually internalize an implicit conceptual data model in the stage after the data has been transformed and loaded into the data warehouse. At this stage, it may be difficult, if not impossible, for analysts or downstream consumers to infer the intended cardinality relationships due to the semantic loss that often occurs during ETL processes. In the absence of explicitly modeled cardinality, relationships that were once nuanced may be flattened or generalized, making incorrect assumptions about the data structure appear valid. While end-user validation is often assumed to be a final safeguard against such misrepresentations, the sheer volume of data in modern analytics environments can mask discrepancies, causing them to be dismissed as statistical noise rather than structural flaws.

Although our framework has primarily positioned probabilistic cardinality notation as a tool for improving the accuracy of analytics, it also plays a critical role in shaping how data is consumed downstream. As shown in the "EduMetrics" case, the approach where the relationship between teacher and class is treated as one-to-many (approach R1) could have been taken due to the downstream applications requiring only the primary teacher for reporting purposes. Alternatively, the business requirement may have been to produce a report with all the associated teachers as an output where the comma-separated format for the list of teachers associated with one class is used instead of repeating the rows for each teacher-class pairing (approach NR1). However, without clear annotation of cardinality expectations the choices may be made inconsistently, leading to diverging interpretations across big data analytics reporting platforms. Explicitly modeling these relationships not only enhances analytical fidelity but also ensures consistency and traceability in data consumption across diverse systems and use cases. Finally, we identify two different approaches to determining thresholds for treating "m": a process driven and a quantitative approach as discussed in Section 4.1.

Our model is complementary to the fuzzy ER model, specifically the second level that deals with fuzzy occurrences of entities and relationships. Based on fuzzy set theory, Zvieli and Chen [74] introduced three levels of fuzziness in the ER model: in the definition of entities and relationships, in their occurrences, and in the values of their attributes. The key difference between fuzzy ER modeling and our proposed probabilistic cardinality approach lies in the focus. Our focus is on the

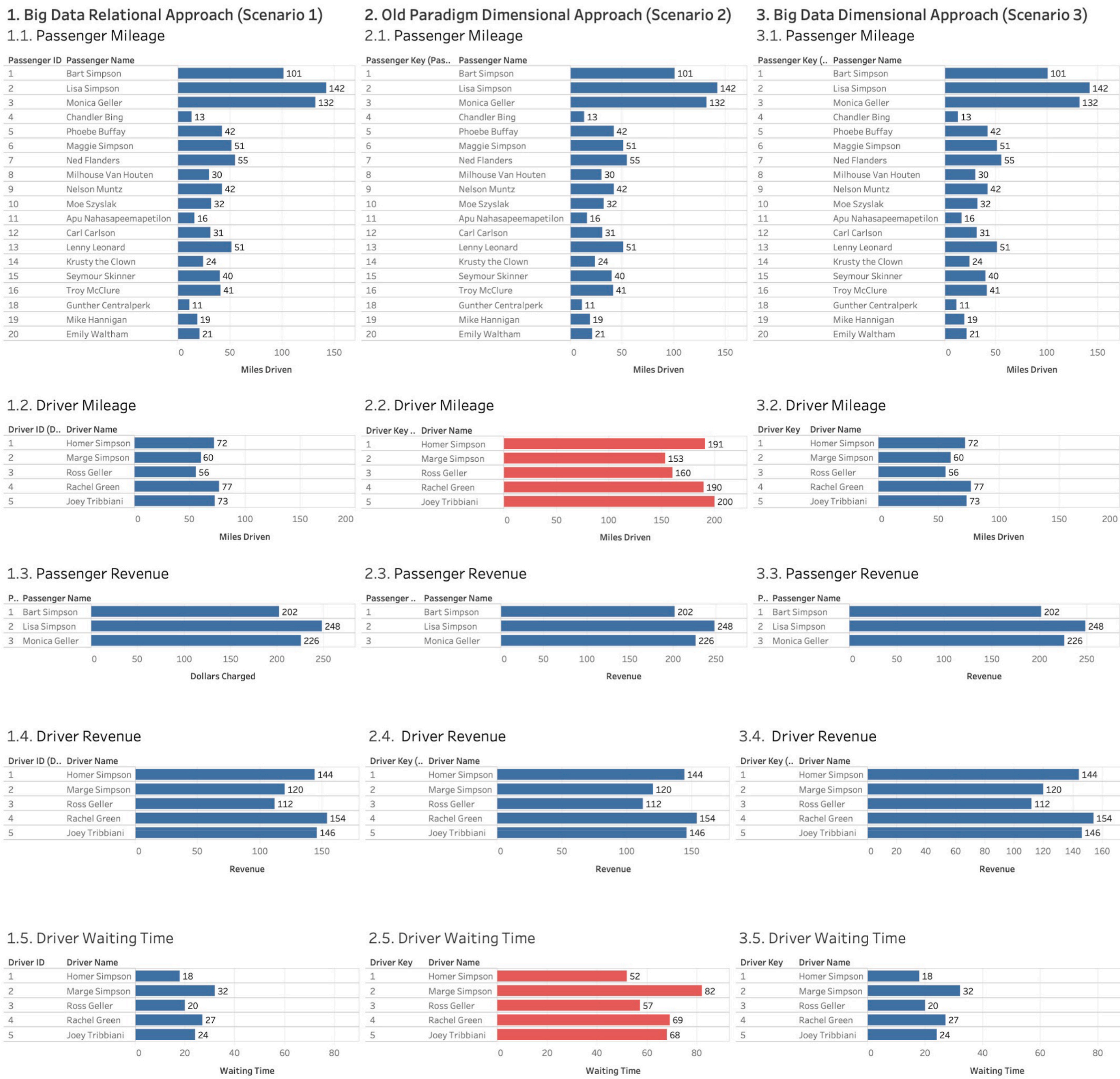


Fig. 13. Tableau dashboard outputs for scenarios 1,2 and 3.

Table 16
Old paradigm and big data dimensional approach results.

Dimensional Approach Scenario Name	Total Number of Relational Tables	Total Number of Dimensions	Total Number of Fact Tables	Total Number of Two-way Metric Aggregations Considered	Total Number of Two-way Incorrect Aggregations	Aggregation Computation Accuracy (%)
Old Paradigm	5	4	1	5	2	60 %
Big Data	5	4	3	5	0	100 %

interpretation of the nature of the relationships in terms of cardinalities for the purpose of downstream analysis while fuzzy ER modeling deals with uncertainty of the existence of instance of each individual relationship either due to lack of complete knowledge or due to the imprecision in interpretation of the nature of the relationship insofar that it results in uncertainty whether particular instance of relationship occurred or not. In other words, our approach does not deal with uncertainties of the data itself but rather addresses the probability of misinterpreting the nature of cardinality of relationships embedded in data on a conceptual level, and consequent design and proper metric allocation.

The distinctions between neighboring paradigms are summarized in

Table 17. The table distinguishes main differences of each paradigm with respect to uncertainty and semantics focus as well as the level of application. The key distinction of our approach compared to other paradigms is that it considers uncertainty in the interpretation of participation and consequent allocation. Additionally, our paradigm is intended to be used on a storage-independent semantic layer, enabling the same probabilistic cardinalities can be expressed in relational, document, or graph ecosystems. Our approach encodes allocation semantics directly at the conceptual modeling layer to support governance and reproducibility. Since the semantic layer is storage type independent, the semantics can be captured as metadata in document-oriented and schema on read environments through configuration files or JSON or YAML descriptors that express expected relationship behavior. This allows probabilistic cardinality to function as a storage independent semantic layer that remains consistent across relational, document-oriented, and graph-based analytical ecosystems.

Table 17
Semantic focus and areas of application of related paradigms.

Paradigm	Uncertainty and Semantics Focus	Area of Application in the Modeling Stack
Fuzzy ER	• Vagueness (degrees of truth). • Relationship is graded (e.g., worksIn = 0.6 means "mostly works in" where worksIn is a relationship between employee and a department). • Extend ER modeling to represent vague concepts and imprecise relationships. • fuzzy set theory	Conceptual Modeling (schema-level vagueness)
Probabilistic Database	• Fact uncertainty (probability of being true). • Relationship is binary but uncertain (exists or not in each world). • Manage uncertain facts where the truth of data is not fully known. (e.g., sensor data as the sensor might not be accurate so it's reading might be true or false) • probability theory	Data/instance layer + query processing: uncertain facts
Graph	• Relationship logic and rules • Inference via graph structure: path discovery, centrality, connectivity patterns. • No uncertainty	Query processing layer: structural queries, path finding, and relationship-centric analytics
Ontology Models	• Ontology semantics capture conceptual meaning of the domain as it exists in reality • describe world-level truths rather than data-level details	Conceptual modeling layer
Probabilistic Cardinality Notation	• Interpretation of participation and consequent allocation uncertainty: probability of multiple allocations • Encodes relationship behavior explicitly at the conceptual level for governance and ETL/ELT clarity. • Probability-based allocation semantics • Storage-independent semantic layer: the same probabilistic cardinalities can be expressed in relational, document, or graph ecosystems.	Conceptual modeling layer (storage-independent semantic layer)

6.3. Metric apportioning notation: replicability

Descriptive analytics are the simplest form of big data analytics methods that use simple statistical methods such as mean, median, mode or frequency measurement [72]. While we are aware that many organizations use more complex metrics derived from more sophisticated calculations, our approach was to focus on the simplest form. Through our example, we showed that even within a context of a simple case of a small dataset, adding one additional data source related to the same transaction can cause inconsistencies in reporting. In other words, we demonstrate the risk in the big data analytics process arising when replicability of metrics is not explicitly documented or modeled. In practice, failure to account for replicability can result in significant misinterpretations, especially when multiple stakeholders with different roles (ex. ETL engineers, data modelers, and analysts) are involved who each rely on their own assumptions regarding data semantics. Traditional data development processes often rely on the assumption that domain-specific nuances will be intuitively recognized by technical professionals through source data inspection. However, as demonstrated in our case studies, such assumptions are dangerous. Two out of five metrics used in our case study presented in Fig. 13 were inconsistently reported for aggregates in Scenario 2. In real-life scenarios, the dashboards are more complex with many of the simple metrics embedded into a variety of complex KPI calculations, making these inconsistencies even harder to identify in the absence of clear semantic agreement among all stakeholders.

In addition, the transformation that is applied in the standard ETL process may obscure original data patterns, making it nearly impossible for analytics developers to retroactively detect misapplied metrics, particularly when such metrics are nuanced. The final line of defense, the end-user validation, is equally unreliable when stakeholders have no baseline expectations, which is illustrated in our example when a wait time metric was introduced. This reinforces the imperative to incorporate replicability considerations into the modeling stage. Documenting metric replicability explicitly not only supports data governance and future maintenance but also enables a shared understanding of replicability by technical and non-technical stakeholders in the big data analytics value chain.

Replicability also interacts with the nature of measures and the type of fact design. For all types of facts (representing additive, semi-additive, and holistic/non-additive measures), as well as for M:N relationships, the treatment of replicability determines how values are allocated or filtered across classifiers. As summarized in Table 18, non-replicable measures must be filtered by a participation condition to ensure unique allocation; replicable measures can appear in full under each classifier without filtering and are suitable for exposure or coverage metrics; and semi-replicable measures may be decomposed into an allocation table (defining weights) and a coverage table to maintain traceability and metric consistency.

As a practical guideline, organizations could embed these rules into software systems for classifying metrics as replicable, non-replicable, or hybrid/semi-replicable, together with the corresponding allocation procedure. For example, waiting time and miles driven facts,

Table 18
Replicability types and respective use in fact table metric treatment.

Replicability Type	Definition/Handling	Appropriate Use
Non-replicable	Must filter via a participation condition to ensure unique allocation.	Allocation-based facts (e.g., transactions, costs).
Replicable	Full value appears under each classifier (no filter applied).	Coverage or exposure metrics.
Semi-replicable	Split into an allocation table (weights) and a coverage table.	Mixed metrics requiring partial allocation and full coverage tracking.

representing exposure metrics, could be treated as replicable across passengers, whereas dollars charged and rating, representing allocation-based metrics should be treated as non-replicable and therefore assigned only to the primary passenger. In hybrid cases, allocation could be segment-based (e.g., when a rider exits midway). We have not developed a formal notation for these cases in this paper. Specifying such rules, especially for hybrid scenarios, remains future work that should be addressed with care.

Classical data warehousing distinguishes three classes of summarizability for measures: additive, semi-additive, and non-additive, a framework most prominently articulated by Kimball and Ross [35]. Summarizability addresses the mathematical behavior of a measure when aggregated across different dimensional hierarchies. The concept of replicability that we introduce operates at a complementary level. Rather than determining whether aggregation is valid, replicability specifies whether a measure may be legitimately copied, shared, or allocated across multiple related entities prior to aggregation. These two perspectives define different but interacting semantic constraints. For example, within our case study, waiting time is both additive and replicable across passengers, while dollars charged is additive but non-replicable and must be assigned to a single participant. Considering summarizability and replicability together produces a matrix of possible combinations that narrows interpretive wiggle room, supports consistent design choices, and improves the fidelity of downstream analytical results. The joint use of both classifications provides a more complete semantic specification than either framework in isolation.

Fig. 14 illustrates the end-to-end workflow, highlighting where probabilistic cardinality notation and replicability decisions are made, who owns them, and how they propagate through the big data analytics value chain and its two complementary implementations. These decisions originate at critical stages denoted in the diagram where allocation semantics are defined and ownership is established between data architects, data modelers, and domain experts. Formally recognizing replicability and probabilistic cardinality notations at these key steps propagates the design decisions enabled by semantics through subsequent schema design and execution, reducing implicit assumptions and preserving metric integrity from data capture to analysis.

7. Conclusion and future work

Overall, recognizing and implementing probabilistic cardinality notation and replicability notation into the data modeling practice of the big data analytics process has the potential to reduce the risk of inconsistent metric reporting. Given the volume, velocity, variety, veracity, and value dimensions of big data, it becomes imperative to model early and often to ensure clarity, alignment, and adaptability throughout the analytics process. Our framework conceptualization of the big data analytics process can serve for future researchers to expand on and contextualize future studies related to specific industries, allowing further exploration of domain-specific adaptations and applications of

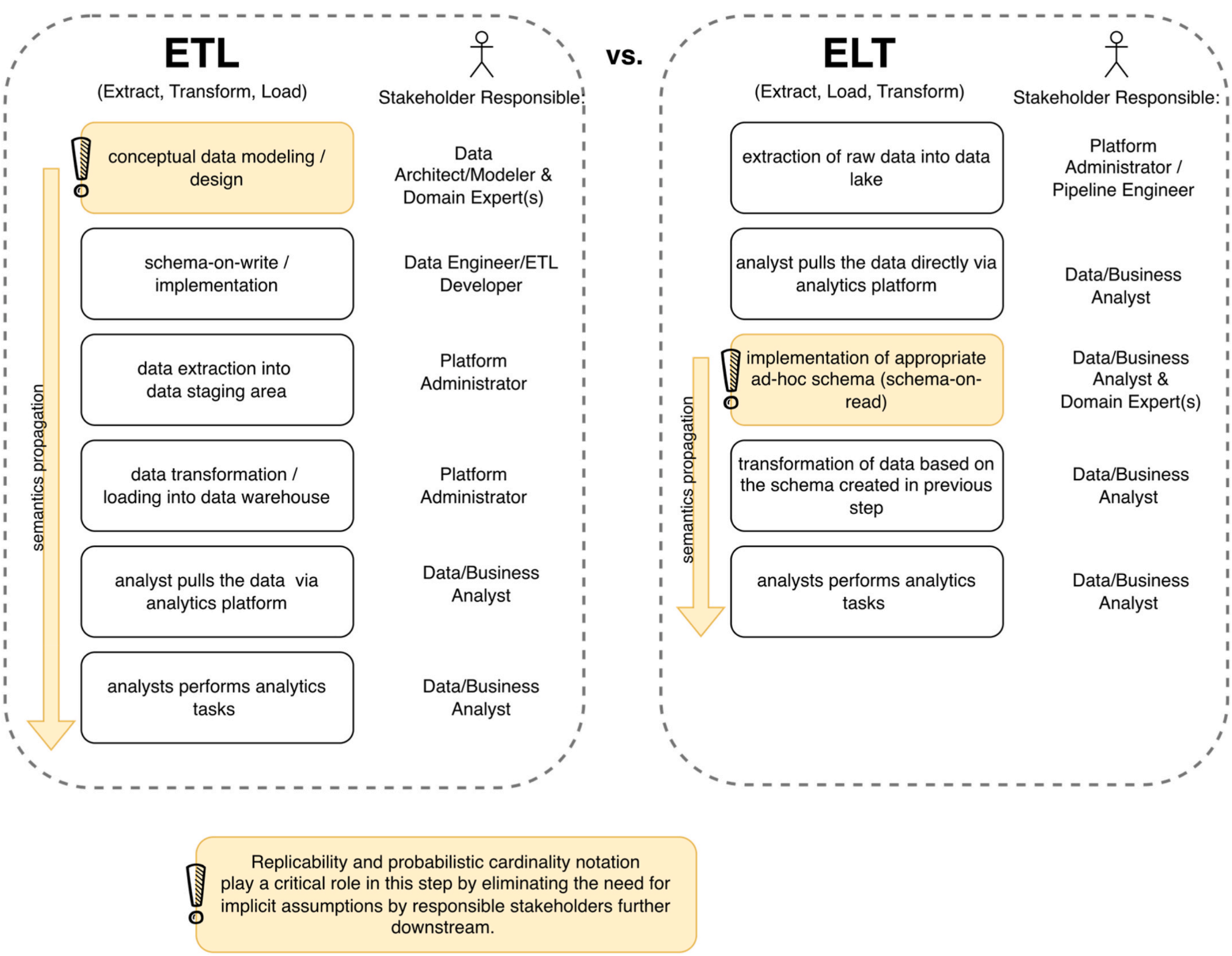


Fig. 14. Propagation of probabilistic cardinality and replicability semantics through ETL and ELT workflows.

the model. Future research can further explore more granular data modeling approaches, such as divisibility of metrics or the introduction of semi-replicable metrics where multiple instances of a classification entity satisfy a given condition, while others do not participate in the relationship. Natural extensions of this work include applying and integrating the proposed concepts into graph databases, document-oriented systems, and other non-relational data stores, where the data modeling contributions can be empirically validated and further refined in real-world settings.

CRediT authorship contribution statement

Jelena Hađina: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Joshua Fogarty:** Visualization, Validation, Investigation. **Boris Jukić:** Writing – review & editing, Supervision, Conceptualization.

Data & code availability

The data, preprocessing scripts and SQL notebooks supporting this study are publicly available at: https://github.com/hadinaj/Array_Submission_Supplemental_Code.

A README in the repository provides instructions for reproducing the case-study tables.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

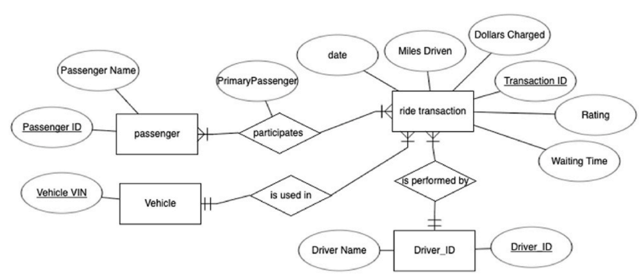
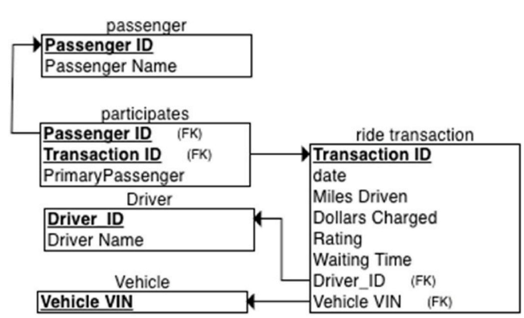
During the preparation of this work the authors used ChatGPT for language editing. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

Table A1
Case Study 2: Data Models for Old Paradigm Dimensional Approach

Data Modeling Approach	Data Model
ER Model	
Relational Model	

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